Answers

1.**A**	2.**B**	3.**C**	4.**C**	5.**C**	6.**D**	7.**D**	8.**B**
9.**D**	10.**B**	11.**A**	12.**C**	13.**A**	14.**A**	15.**B**	16.**B**
17.**E**	18.**D**	19.**D**	20.**C**	21.**A**	22.**A**	23.**C**	24.**B**
25.**A**	26.**D**	27.**B**	28.**B**	29.**C**	30.**D**	31.**A**	32.**D**
33.**A**	34.**D**	35.**D**	36.**C**	37.**B**	38.**C**	39.**D**	40.**C**
41.**C**	42.**B**	43.**D**	44.**C**	45.**C**	46.**C**	47.**B**	48.**C**
49.**C**	50.**C**	51.**C**	52.**D**	53.**B**	54.**B**	55.**D**	56.**B**
57.**A**	58.**B**	59.**C**	60.**A**	61.**C**	62.**B**	63.**A**	64.**C**
65.**A**	66.**D**	67.**B**	68.**4**	69.**A**	70.**D**	71.**A**	72.**D**
73.**D**	74.**19**	75.**3**	76.**D**	77.**7**	78.**4**	79.**B**	80.**B**
81.**A**	82.**D**	83.**C**	84.**D**	85.**C**	86.**4**	87.**3**	88.**A**
89.**C**	90.**3**	91.**D**	92.**21**	93.**A**	94.**C**	95.**B**	96.**D**
97.**D**	98.**D**	99.**A**	100.**D**	101.**D**	102.**C**	103.**A**	104.**C**
105.**A**	106.**A**	107.**3**	108.**B**	109.**A**	110.**D**	111.**3**	112.**D**
113.**C**	114.**D**	115.**D**	116.**6**	117.**6**	118.**9**	119.**2**	120.**B**
121.**A**							

Explanations

1.**A**

From the given statements, we can infer that the amounts spent by the women are Rs.2234, Rs.1193, Rs.1340, and Rs.2517. Also, we know that one of the 5 women spent Rs.1378 more than Chellamma. Also, we know that each woman spent at least Rs.1000.

The person who spent Rs.2234 cannot be the person who spent Rs.1378 more than Chellamma. Also, we know that Chellamma spent the least among the five women.

Case 1:
Chellamma spent Rs.1193.
=> One of the 5 women spent 1193+1378 = Rs.2571.

We know that Shahnaz spent the most. Therefore, Shahnaz should have spent Rs.2571.

Archana didn't spend Rs.2517. Also, Helen spent more than Dhenuka.
Therefore, Helen should have spent Rs.2517.

Dhenuka didn't spend Rs.1340.
Therefore, Dhenuka should have spent Rs. 2234 and Archana should have spent Rs.1340.

However, it has been given that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193.
According to the given order, Archana arrived before Dhenuka. Therefore, we can eliminate this case.

Case 2:

Rs.2517 is the highest amount spent. Shahnaz spent Rs.2517.
=> Amount spent by Chellamma = 2517 - 1378 = Rs.1139

Dhenuka didn't spend Rs.1340. Helen spent more than Dhenuka. Therefore, Dhenuka should have spent

Rs.1193.

Now, we know that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. Helen did not arrive before Dhenuka but Archana did. Therefore, Archana should have spent Rs.2234 and Helen should have spent Rs.1340.

According to given conditions amount spent by everyone is,

ARCHANA	CHELLAMMA	DHENUKA	HELEN	SHAHNAZ
2234	1139	1193	1340	2517
	=2517-1378			

Hence, option A is the right answer.

VIEW SOLUTION

2. **B**
From the given statements, we can infer that the amounts spent by the women are Rs.2234, Rs.1193, Rs.1340, and Rs.2517. Also, we know that one of the 5 women spent Rs.1378 more than Chellamma. Also, we know that each woman spent at least Rs.1000.

The person who spent Rs.2234 cannot be the person who spent Rs.1378 more than Chellamma. Also, we know that Chellamma spent the least among the five women.

Case 1:
Chellamma spent Rs.1193.
=> One of the 5 women spent 1193+1378 = Rs.2571.

We know that Shahnaz spent the most. Therefore, Shahnaz should have spent Rs.2571.

Archana didn't spend Rs.2517. Also, Helen spent more than Dhenuka.
Therefore, Helen should have spent Rs.2517.

Dhenuka didn't spend Rs.1340.
Therefore, Dhenuka should have spent Rs. 2234 and Archana should have spent Rs.1340.

However, it has been given that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. According to the given order, Archana arrived before Dhenuka. Therefore, we can eliminate this case.

Case 2:

Rs.2517 is the highest amount spent. Shahnaz spent Rs.2517.
=> Amount spent by Chellamma = 2517 - 1378 = Rs.1139

Dhenuka didn't spend Rs.1340. Helen spent more than Dhenuka. Therefore, Dhenuka should have spent Rs.1193.

Now, we know that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. Helen did not arrive before Dhenuka but Archana did. Therefore, Archana should have spent Rs.2234 and Helen should have spent Rs.1340.

According to given conditions amount spent by everyone is,

ARCHANA	CHELLAMMA	DHENUKA	HELEN	SHAHNAZ
2234	1139	1193	1340	2517
	=2517-1378			

Hence, option B is the right answer.

VIEW SOLUTION

3. **C**

From the given statements, we can infer that the amounts spent by the women are Rs.2234, Rs.1193, Rs.1340, and Rs.2517. Also, we know that one of the 5 women spent Rs.1378 more than Chellamma. Also, we know that each woman spent at least Rs.1000.

The person who spent Rs.2234 cannot be the person who spent Rs.1378 more than Chellamma. Also, we know that Chellamma spent the least among the five women.

Case 1:
Chellamma spent Rs.1193.
=> One of the 5 women spent 1193+1378 = Rs.2571.

We know that Shahnaz spent the most. Therefore, Shahnaz should have spent Rs.2571.

Archana didn't spend Rs.2517. Also, Helen spent more than Dhenuka.
Therefore, Helen should have spent Rs.2517.

Dhenuka didn't spend Rs.1340.
Therefore, Dhenuka should have spent Rs. 2234 and Archana should have spent Rs.1340.

However, it has been given that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. According to the given order, Archana arrived before Dhenuka. Therefore, we can eliminate this case.

Case 2:

Rs.2517 is the highest amount spent. Shahnaz spent Rs.2517.
=> Amount spent by Chellamma = 2517 - 1378 = Rs.1139

Dhenuka didn't spend Rs.1340. Helen spent more than Dhenuka. Therefore, Dhenuka should have spent Rs.1193.

Now, we know that the woman who spent Rs. 2234 arrived before the lady who spent Rs. 1193. Helen did not arrive before Dhenuka but Archana did. Therefore, Archana should have spent Rs.2234 and Helen should have spent Rs.1340.

According to given conditions amount spent by everyone is,

ARCHANA	CHELLAMMA	DHENUKA	HELEN	SHAHNAZ
2234	1139	1193	1340	2517
	=2517-1378			

Hence, option C is the right answer.

VIEW SOLUTION

4. **C**

We are given that A and G sit to the extreme right on the dias, but the order is not mentioned. So, seats 6 and 7 must be A and G in some order.

B sits in the middle of the presentation, occupying seat number 4.

C and D have to be seated as far as possible, and seats 4, 6 and 7 are already occupied. So, C and D must be placed in seats 1 and 5 in some order.

The seats left are 2 and 3, and E and F have to be placed in those seats in some order.

Putting all the information in the table, we get,

LEFT						RIGHT
SEAT-1	SEAT-2	SEAT-3	SEAT-4	SEAT-5	SEAT-6	SEAT-7
C /D	E/F	F/E	B	D/C	G/A	A/G

All the combinations mentioned in options A, B and D are possible from the above configuration. However, the combination mentioned in option C is not possible as E and G cannot be seated on either side of B.

Hence, the correct answer is option C.

VIEW SOLUTION

5. **C**

We are given that A and G sit to the extreme right on the dias, but the order is not mentioned. So, seats 6 and 7 must be A and G in some order.

B sits in the middle of the presentation, occupying seat number 4.

C and D have to be seated as far as possible, and seats 4, 6 and 7 are already occupied. So, C and D must be placed in seats 1 and 5 in some order.

The seats left are 2 and 3, and E and F have to be placed in those seats in some order.

Putting all the information in the table, we get,

LEFT						RIGHT
SEAT-1	SEAT-2	SEAT-3	SEAT-4	SEAT-5	SEAT-6	SEAT-7
C /D	E/F	F/E	B	D/C	G/A	A/G

We can see that A, C, D or G can be placed in the corner. But F cannot be seated at either end.

Hence, the correct answer is option C.

VIEW SOLUTION

6. **D**

We are given that A and G sit to the extreme right on the dias, but the order is not mentioned. So, seats 6 and 7 must be A and G in some order.

B sits in the middle of the presentation, occupying seat number 4.

C and D have to be seated as far as possible, and seats 4, 6 and 7 are already occupied. So, C and D must be placed in seats 1 and 5 in some order.

The seats left are 2 and 3, and E and F have to be placed in those seats in some order.

Putting all the information in the table, we get,

LEFT						RIGHT
SEAT-1	SEAT-2	SEAT-3	SEAT-4	SEAT-5	SEAT-6	SEAT-7
C /D	E/F	F/E	B	D/C	G/A	A/G

We can clearly see that E and A cannot be seated together.

Hence, the correct answer is option D.

VIEW SOLUTION

7. **D**

B says that if D gets power or telecom then he must get the other one.Option D clearly violates that.

VIEW SOLUTION

8. **B**
Since C wants either home or finance or none so options A and D are eliminated.
Since F does not want any portfolio if D gets one, Option C is eliminated.

VIEW SOLUTION

9. **D**
We are given that there are two housewives, one professor, one engineer, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an engineer or an accountant.

C is an accountant, has to be male, and is married to F, who is a professor, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Professor	Housewife	Engineer
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to the housewife. We know that there are only two married couples in the group, and C and F are already one couple in that group. We also know that D is a housewife and married to a lawyer, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and engineer professions. Given that E is not a housewife, E has to be the Engineer and B has to be the housewife. Since the engineer cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Professor	Housewife	Engineer
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

From the table, we can see that A and D are a married couple.

Hence, the correct answer is option D.

VIEW SOLUTION

10. **B**
We are given that there are two housewives, one professor, one engineer, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an engineer or an accountant.

C is an accountant, has to be male, and is married to F, who is a professor, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Professor	Housewife	Engineer
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to the housewife. We know that there are only two married couples in the group, and C and F are already one couple in that group. We also know that D is a housewife and married to a lawyer, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and engineer professions. Given that E is not a housewife, E has to be the Engineer and B has to be the housewife. Since the engineer cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Professor	Housewife	Engineer
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

We can see that A, C and E are the three males in the group.

Hence, the correct answer is option B.

VIEW SOLUTION

11. **A**
We are given that there are two housewives, one professor, one engineer, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an engineer or an accountant.

C is an accountant, has to be male, and is married to F, who is a professor, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Professor	Housewife	Engineer
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to the housewife. We know that there are only two married couples in the group, and C and F are already one couple in that group. We also know that D is a housewife and married to a lawyer, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and engineer professions. Given that E is not a housewife, E has to be the Engineer and B has to be the housewife. Since the engineer cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Professor	Housewife	Engineer
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

We can see that E is an engineer.

Hence, the correct answer is option A.

VIEW SOLUTION

12. **C**

For option A, put n_1 = 1, then n_2 will be 2 , hence equation won't hold true.
Now for option B, if we say M = 5 , then n_1 will be 0 and n_2 will be 1, hence equation won't hold true.
And equation in option D will not hold true too as put M = 5 , then n_1 will be 0 and n_2 will be 1.
Now in option C, equation is valid as put n_1 = 1 , then n_2 will be 2. Which is holding the equation valid too.
Hence our answer will be C.

VIDEO SOLUTION

13. **A**

In option A, n_1 >=5 which is before instruction value so after going 5 backward and 5 forward value of n_2 will remain equal to n_1

So our answer will be A as other options have insufficient information

VIDEO SOLUTION

14. **A**

Option B and C are irrelevant and in option D, n_1 should be 0
Hence option A will be our answer as n_1 is the value before instruction as 0 and n_2 is value after instructions which is also 0.

VIDEO SOLUTION

15. **B**

Let's consider each statement one by one

A) Suppose M=50, n_1=5

In this case, n_2 = 5 + 10 + 10 = 25. Hence, $n_2 - n_1 = 20$ even though $n_1 \neq 0$. Hence, statement A is false.

B) Suppose M=21 and n_1=1

In this case, n_2 = 1 + 10 + 10 = 21. Hence, $n_2 - n_1 = 20$. Thus, if M>20 then this statement is true.

If M = 19

In this case, the process stops when M is reached. Hence, n_2=19 and $n_2 - n_1 = 18$. Hence, option B is true.

C) As we have seen earlier, $n_2 - n_1 = 20$ in this case. Hence, option C is false.

D) Suppose $n_1 = M$. In this case, $n_2 = n_1$. Hence, option D is false.

VIDEO SOLUTION

16. **B**

Before directly trying to answer the question, it is important to gather all the information given by the question.

There are three houses on each side of the road => Draw 6 lines, 3 in each row, to accommodate P, Q, R, S, T and U.

The houses are of different colours and different heights.

T is tallest and is opposite to red house => Let's number T as 1.

Shortest house is opposite to green house.

U is orange and is between P and S => Two cases arise here. P-U-S is one possibility and the other possibility is S-U-P.

P	**U**	**S**
R	**Q**	**T**

R is yellow and is opposite to P.

Q is green and is opposite to U. We know that green house is opposite to the shortest house. This implies that U is the shortest house => Number of U is 6.

P is white and is taller than R but shorter than S and Q => Apart from T, S and Q are also taller than P => S and Q can be 2 and 3 in any order => Number of P is 4 and number of R is 5.

Name	Colour	Height-Rank
P	White	4
Q	Green	2 or 3
R	Yellow	5
S	Red	3 or 2
T	Blue	1
U	Orange	6

We know that P is opposite to R and Q is opposite to U => S is opposite to T

It is given that T is opposite to red house => S is the red house and hence T is the blue house.

T is the tallest house and hence the colour of the tallest house is blue.

VIEW SOLUTION

17. **E**

Before directly trying to answer the question, it is important to gather all the information given by the question.

There are three houses on each side of the road => Draw 6 lines, 3 in each row, to accommodate P, Q, R, S, T and U.

The houses are of different colours and different heights.

T is tallest and is opposite to red house => Let's number T as 1.

Shortest house is opposite to green house.

U is orange and is between P and S => Two cases arise here. P-U-S is one possibility and the other possibility is S-U-P.

P	**U**	**S**
R	**Q**	**T**

R is yellow and is opposite to P.

Q is green and is opposite to U. We know that green house is opposite to the shortest house. This implies that U is the shortest house => Number of U is 6.

P is white and is taller than R but shorter than S and Q => Apart from T, S and Q are also taller than P => S and Q can be 2 and 3 in any order => Number of P is 4 and number of R is 5.

Name	Colour	Height-Rank
P	White	4
Q	Green	2 or 3
R	Yellow	5
S	Red	3 or 2
T	Blue	1
U	Orange	6

We only know that the second tallest house is either Q or S. Hence the answer is cannot be determined.

VIEW SOLUTION

18. **D**

Before directly trying to answer the question, it is important to gather all the information given by the question.

There are three houses on each side of the road => Draw 6 lines, 3 in each row, to accommodate P, Q, R, S, T and U.

The houses are of different colours and different heights.

T is tallest and is opposite to red house => Let's number T as 1.

Shortest house is opposite to green house.

U is orange and is between P and S => Two cases arise here. P-U-S is one possibility and the other possibility is S-U-P.

P	**U**	**S**
R	**Q**	**T**

R is yellow and is opposite to P.

Q is green and is opposite to U. We know that green house is opposite to the shortest house. This implies that U is the shortest house => Number of U is 6.

P is white and is taller than R but shorter than S and Q => Apart from T, S and Q are also taller than P => S and Q can be 2 and 3 in any order => Number of P is 4 and number of R is 5.

Name	Colour	Height-Rank
P	White	4
Q	Green	2 or 3
R	Yellow	5
S	Red	3 or 2
T	Blue	1
U	Orange	6

We know that P is opposite to R and Q is opposite to U => S is opposite to T

It is given that T is opposite to red house => S is the red house and hence T is the blue house.

So, we know the colours of all houses and heights of P, R, T and U.

In this question, we are asked to find the house that is opposite to yellow house. R is the yellow house, P is opposite to R and S is on the other corner in P's row. Hence S is the house that is diagonally opposite to yellow house and the colour of S is Red.

VIEW SOLUTION

19. **D**

We know that $P \cap A = \phi$ means that the pluto is not an alsatian and $P \cup A = D$ means that the pluto along with alsatian makes the set D . hence option D - Both A and C.

VIEW SOLUTION

20. **C**

We know that all dogs are vertebrates. Hence $D \cap V)$ = D . Ahead it is given that $Y = F \cap D$ is not a null set which implies that some fish are dogs. Hence , option C.

VIEW SOLUTION

21. **A**

It is given that $X = M \cap D$ is such that $X = D$, which means D is a subset of M . Which means all dogs are mammals. Hence , option A.

VIEW SOLUTION

22. **A**

From $P \cap D$ is a set containing just pluto ,as pluto is subset of dog D, hence $P \cap D$ = P . Also $P \cup M$ would be set of pluto and all mammals together.

VIEW SOLUTION

23. **C**

E = 3Y
Z = W/2
Y > Z

Using statement A alone, Z and W can be determined but not E

Using statement B, Y = W. So, using this statement alone, we can get a relation between all the four ages but cannot determine the absolute values.

Using both the statements, we can determine the value of E.

10 = W/2

W = 20 = Y

E = 3*20 = 60

So, c) is the correct answer.

VIEW SOLUTION

24. **B**
E = 3Y
Z = W/2
Y > Z
So, Y > W/2 => 2Y > W => 2E/3 > W => E > 3W/2 => E > W
So, Elle is older than W

VIEW SOLUTION

25. **A**
The arrangement of people is as follows: C/D F/E E/F B D/C A/G G/A

So, E and A can never sit together.

VIEW SOLUTION

26. **D**
_ _ _ B _ _ _

A and G will sit in 6th and 7th position from left in any order.

C and D must be as far apart as possible. Hence, they'll sit in 1st and 5th positions from left in any order.

Hence, only A, C, D and G can sit in the corners.

Among, the given options F cannot sit in the extreme end.

VIEW SOLUTION

27. **B**
We are given that there are two housewives, one lecturer, one architect, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an architect or an accountant.

C is an accountant, has to be male, and is married to F, who is a lecturer, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Lecturer	Housewife	Architect
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to D, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and architect professions. Given that E is not a housewife, E has to be the Architect and B has to be the housewife. Since the Architect cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Lecturer	Housewife	Architect
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

From the table, we can see that E is an Architect.

Hence, the correct answer is option B.

VIEW SOLUTION

28. **B**

We are given that there are two housewives, one lecturer, one architect, one lawyer and one accountant among A, B, C, D, E and F.

The other information given is that only two married couples are in the group.

We are given that the lawyer is married to D, and D is a housewife.

We are also given that no woman in the group is either an architect or an accountant.

C is an accountant, has to be male, and is married to F, who is a lecturer, who has to be female.

Putting this information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Lecturer	Housewife	Architect
NAME		D	C	F		
GENDER	Male	Female	Male	Female		
MARRIED TO	D		F	C		

We are also given that A is married to D, so this has to be the second couple, and A has to be the lawyer.

This leaves us with B and E, as well as housewife and architect professions. Given that E is not a housewife, E has to be the Architect and B has to be the housewife. Since the Architect cannot be a woman, E has to be a man, and B has to be a woman.

Putting all the information in the table, we get,

PROFESSION	Lawyer	Housewife	Accountat	Lecturer	Housewife	Architect
NAME	A	D	C	F	B	E
GENDER	Male	Female	Male	Female	Female	Male
MARRIED TO	D	A	F	C		

From the table, we can see 3 males in the group.

Hence, the correct answer is option B.

VIEW SOLUTION

29. **C**

From the table we can say that number of students who opted for E2 after reshuffle = 5 + 34 + 6 + 3 + 5 + 7 + 16 = 76.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = 76 - 30 = 46.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = 46 - 10 = 36.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = 9 + 5 + 10 + 1 + 4 + 2 = 31.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = 31 - 6 = 25.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = 25 - 2 = 23.

All the students from E7 moved to one of electives among E1 to E6.
Therefore, the number of students who were enrolled in E7 before reshuffle = 4 + 16 + 30 + 5 + 5 + 41 = 101.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E7 before reshuffle = 300 - (46+36+31+25+23+101) = 38.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25 i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.
As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
	Total	18	76	79	21	45	61	300

From the table we can see that the number of students who enrolled for E1 and E4 decreased from 31 and 25 to 18 and 21 respectively.

Therefore, option C is the correct answer.

VIDEO SOLUTION

30. **D**
From the table we can say that number of students who opted for E2 after reshuffle = 5 + 34 + 6 + 3 + 5 + 7 + 16 = 76.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = 76 - 30 = 46.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = 46 - 10 = 36.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = 9 + 5 + 10 + 1 + 4 + 2 = 31.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = 31 - 6 = 25.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = 25 - 2 = 23.

All the students from E7 moved to one of electives among E1 to E6.
Therefore, the number of students who were enrolled in E7 before reshuffle = 4 + 16 + 30 + 5 + 5 + 41 = 101.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E7 before reshuffle = 300 - (46+36+31+25+23+101) = 38.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46 i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.
As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
	Total	18	76	79	21	45	61	300

Form the table, we can see that after the reshuffle the number of students in electives E1 to E6 are 18, 76, 79, 21, 45 and 61 in that order.

Therefore, option D is the correct answer.

VIDEO SOLUTION

31. **A**

From the table we can say that number of students who opted for E2 after reshuffle = 5 + 34 + 6 + 3 + 5 + 7 + 16 = 76.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = 76 - 30 = 46.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = 46 - 10 = 36.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = 9 + 5 + 10 + 1 + 4 + 2 = 31.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = 31 - 6 = 25.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = 25 - 2 = 23.

All the students from E7 moved to one of electives among E1 to E6.
Therefore, the number of students who were enrolled in E7 before reshuffle = 4 + 16 + 30 + 5 + 5 + 41 = 101.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E7 before reshuffle = 300 - (46+36+31+25+23+101) = 38.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

From Elective		To Elective						
		E1	E2	E3	E4	E5	E6	Total
	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.
As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

From Elective		To Elective						
		E1	E2	E3	E4	E5	E6	Total
	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

From Elective		To Elective						
		E1	E2	E3	E4	E5	E6	Total
	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
	Total	18	76	79	21	45	61	300

We can see from the table that number of students enrolled in E1 dropped to 18. Hence, all the students who moved from E1 to any other elective will have to re-enroll in E1.

We can see that the number of students who enrolled for E1 prior to reshuffle = 31. Out of these 31 students, 9 students didn't move to any other elective whereas remaining 22 students moved to other electives. Hence, all these 22 students have to re-enroll in E1.

Therefore, the total number of students in E1 post re-enrollment = 18 + 22 = 40 which is shown in the table.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	31	0	0	0	0	0	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
	Total	40	71	69	20	41	59	300

Therefore, the sequence of electives in decreasing order of their final enrollments = E2, E3, E6, E5, E1, E4.

Hence, option A is the correct answer.

VIDEO SOLUTION

32. **D**

From the table we can say that number of students who opted for E2 after reshuffle = 5 + 34 + 6 + 3 + 5 + 7 + 16 = 76.

It is given us that the number of students in E2 increased by 30 after the change process. Hence, we can say that the number of students who were enrolled in E2 before reshuffle = 76 - 30 = 46.

It is given that before the change process there were 10 more students in E2 than in E3. Therefore, the number of students who were enrolled in E3 before reshuffle = 46 - 10 = 36.

Number of students who moved from E1 to all other electives are known. Therefore, the number of students who were enrolled in E1 before reshuffle = 9 + 5 + 10 + 1 + 4 + 2 = 31.

It is given that before the change process there were 6 more students in E1 than in E4. Therefore, the number of students who were enrolled in E4 before reshuffle = 31 - 6 = 25.

Also, it is given that E4 had 2 more students than E6 before reshuffle. Therefore, the number of students who were enrolled in E6 before reshuffle = 25 - 2 = 23.

All the students from E7 moved to one of electives among E1 to E6.
Therefore, the number of students who were enrolled in E7 before reshuffle = 4 + 16 + 30 + 5 + 5 + 41 = 101.

Except E5 we know the number of students who were enrolled in all electives. We also know that there were total 300 students who opted for exactly 1 elective.

Hence, the the number of students who were enrolled in E7 before reshuffle = 300 - (46+36+31+25+23+101) = 38.

For each elective, the number of students who were enrolled before reshuffle will be same as sum of the number of students who moved from that elective to another elective including no movement cases.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2		34	8		2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E2,

Number of students who moved to E1 + 34 + 8 + Number of students who moved to E4 + 2 + 2 = 46

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 0

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4		3	2	14		4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E4,

Number of students who moved to E1 + 3 + 2 + 14 + Number of students who moved to E5 + 4 = 25

i.e. Number of students who moved from to E1 = Number of students who moved from to E5 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6		7	3		2	9	23
	E7	4	16	30	5	5	41	101

For elective E6,

Number of students who moved to E1 + 7 + 3 + Number of students who moved to E4 + 2 + 9 = 23

i.e. Number of students who moved from to E1 = Number of students who moved from to E4 = 1 {As the remaining blanks can be filled by either 0 or 1}

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25			2	36
	E4	1	3	2	14	1	4	25
	E5		5			30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

It is given that after the reshuffle, the number of students in E4 was 3 more than that in E1.
As of now the number of students enrolled in E4 after reshuffle = 1 + 0 + E3 to E4 + 14 + E5 to E4 + 1 + 5 = 21 + {E3 to E4} + {E5 to E4}

Also, the number of students enrolled in E1 after reshuffle = 9 + 0 + 2 + 1 + E5 to E1 + 1 + 4 = 17 + E5 to E1.

Hence, it is possible only when E5 to E1 = 1 and E3 to E4 = E5 to E4 = 0.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0		2	36
	E4	1	3	2	14	1	4	25
	E5	1	5		0	30		38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101

Remaining blank places can be filled easily as we know the total sum of each row.

Therefore, the number of students who moved from E3 to E5 = the number of students who moved from E5 to E3 = the number of students who moved from E5 to E6 = 1.

		To Elective						
		E1	E2	E3	E4	E5	E6	Total
From Elective	E1	9	5	10	1	4	2	31
	E2	0	34	8	0	2	2	46
	E3	2	6	25	0	1	2	36
	E4	1	3	2	14	1	4	25
	E5	1	5	1	0	30	1	38
	E6	1	7	3	1	2	9	23
	E7	4	16	30	5	5	41	101
	Total	18	76	79	21	45	61	300

We are asked the largest change in its enrollment as a percentage of its original enrollment for all 6 electives but as we can see there are only 4 electives. Hence, we will check only for E1, E2, E3 and E6.

The percentage change in the number of students for E1 = $\frac{18-31}{31} \times 100 \approx 42$ %

The percentage change in the number of students for E2 = $\frac{76-46}{46} \times 100 \approx 65$ %

The percentage change in the number of students for E3 = $\frac{79-36}{36} \times 100 \approx 119$ %

The percentage change in the number of students for E6 = $\frac{61-23}{23} \times 100 \approx 165$ %

We can see that the percent change in the number of student for E6 is the largest. Therefore, option D is the correct answer.

VIDEO SOLUTION

33. **A**

The total number of biscuits = 5, the total number of candies =3 and the total number of savouries = 12-(3+5)=4

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D,E,F and K are savouries and placed at 13,14,15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C,H are candies and I,J,L are biscuits. It is given that A, B are place consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I,J,L,A,B are biscuits and C,H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	L	A	B	I/J	J/I	___	___	C	H/G	G/H	___	D	E	F	K
Item Type	__	B	B	B	B	B	___	___	C	C	C	___	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	__	C	H/G	G/H	__	L	A	B	I/J	J/I	___	D	E	F	K
Item Type	__	__	C	C	C	___	B	B	B	B	B	___	S	S	S	S

The number of arrangements for the first case = 2*2=4

The number of arrangements for the second case = 2*2=4

The total number of arrangements = 4+4=8

VIDEO SOLUTION

34. **D**

The total number of biscuits = 5, the total number of candies =3 and the total number of savouries = 12-(3+5)=4

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D,E,F and K are savouries and placed at 13,14,15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C,H are candies and I,J,L are biscuits. It is given that A, B are place consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I,J,L,A,B are biscuits and C,H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	L	A	B	I/J	J/I	___	___	C	H/G	G/H	___	D	E	F	K
Item Type	__	B	B	B	B	B	___	___	C	C	C	___	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	__	C	H/G	G/H	__	L	A	B	I/J	J/I	___	D	E	F	K
Item Type	__	__	C	C	C	___	B	B	B	B	B	___	S	S	S	S

Option A and C are wrong as candies can come before biscuits and vice versa. B is not necessarily true as there can be one empty shelf too as shown in the table. Option D is true as there are at least 4 shelves between B and C. Hence D is the answer.

VIDEO SOLUTION

35. **D**

The total number of biscuits = 5, the total number of candies =3 and the total number of savouries = 12-(3+5)=4

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D,E,F and K are savouries and placed at 13,14,15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C,H are candies and I,J,L are biscuits. It is given that A, B are place consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I,J,L,A,B are biscuits and C,H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	L	A	B	I/J	J/I	___	___	C	H/G	G/H	___	D	E	F	K
Item Type	__	B	B	B	B	B	___	___	C	C	C	___	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	__	C	H/G	G/H	__	L	A	B	I/J	J/I	___	D	E	F	K
Item Type	__	__	C	C	C	___	B	B	B	B	B	___	S	S	S	S

G is a candy. Hence D is the answer.

VIDEO SOLUTION

36. **C**

The total number of biscuits = 5, the total number of candies =3 and the total number of savouries = 12-(3+5)=4

Representing the candies as C, biscuits as B and savories as S. K is to be placed in shelf number 16. D, E and F are savouries and are to be placed in consecutively numbered shelves in increasing order after all the biscuits and candies. Since there is no empty shelf between the items of same type, D,E,F and K are savouries and placed at 13,14,15 and 16 respectively. This can be tabulated as follows:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code													D	E	F	K
Item Type													S	S	S	S

The shelf 12 will be empty.

It is given that items are to be placed such that all items of same type are clustered together.

From 1, A and B are to be placed in consecutively numbered shelves in increasing order.

From 6, C is a candy and is to be placed in a shelf preceded by two empty shelves and from 7, L is to be placed in a shelf preceded by exactly one empty shelf.

Hence C and L are items of different types. Since C is a candy, L will be a biscuit.

From 5, L and J are items of the same type, while H is an item of a different type.

Since I and J are clustered together, I, J and L are biscuits and H is a candy.

So C,H are candies and I,J,L are biscuits. It is given that A, B are place consecutively. Hence A and B are items of same types. So A, B should be biscuits because if they are candies, there will be 4 candies.

Hence, I,J,L,A,B are biscuits and C,H and G are candies.

Now there are two empty shelves before C and exactly one empty shelf before L, then the different cases can be tabulated as follows:

Case 1:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	L	A	B	I/J	J/I	___	___	C	H/G	G/H	___	D	E	F	K
Item Type	__	B	B	B	B	B	___	___	C	C	C	___	S	S	S	S

Case 2:

Shelf No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Item Code	__	__	C	H/G	G/H	__	L	A	B	I/J	J/I	___	D	E	F	K
Item Type	__	__	C	C	C	___	B	B	B	B	B	___	S	S	S	S

From the table(case 2), only 1,2,6 and 12 are empty in the same arrangement. Hence, C is the answer.

VIDEO SOLUTION

37. **B**

After first instruction: A = 5-2 = 3L. C = 2L. In the third instruction, A yields 2L to C. It means the capacity of A before third operation was 3L. It means that C's water was either drained or transferred to B. Option B is correct.

VIEW SOLUTION

38. **C**

We know that after 3 operations A contain 1 ltr and 4 th operation is Drain(a) , so 1 ltr is drained and only 4 ltrs remain in all there cylinders. Also after 6 operations A should have all the 4 ltrs in it. hence the other 2 vessels must contain 0 ltrs because there is only one drain operation in the entire sequence.

VIEW SOLUTION

39. **D**

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

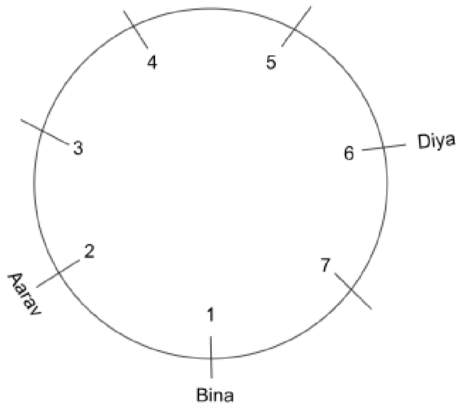
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

1) Passing the buck second to the right in round 4 and second to the right in round 5.
2) Passing the buck second to the left in round 4 and to the left in round 5.
3) Passing the buck to the left in round 4 and second to the left in round 5.

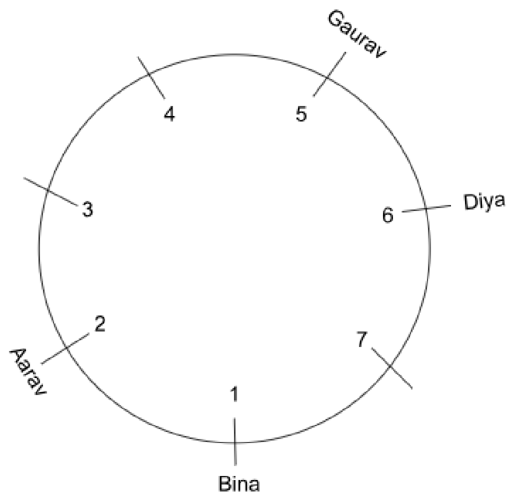
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.

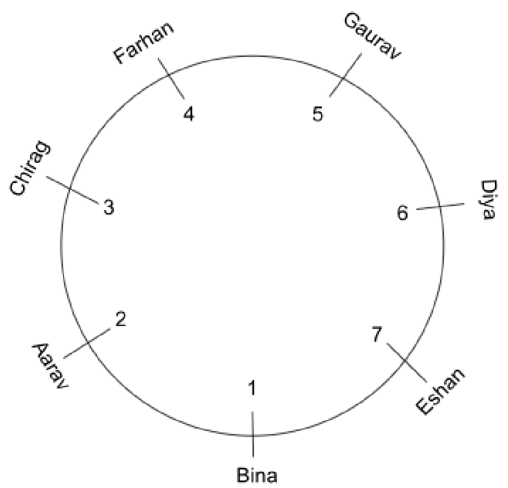


Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.
If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.
The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



In rounds 4 and 5, the possibilities were

1) Passing the buck second to the right in round 4 and second to the right in round 5. Here, the buck would have gone to Farhan in round 4.
2) Passing the buck second to the left in round 4 and to the left in round 5. Here, the buck would have gone to Bina in round 4.
3) Passing the buck to the left in round 4 and second to the left in round 5. Here, the buck would have gone to Eshan in round 4.

In round 4, the buck might have gone to either Farhan, Bina or Eshan, which is not known to us. Therefore, the count of the number of times the buck was received by them cannot be determined uniquely, whereas it can be uniquely determined in the case of Gaurav, which is 1 in round 7.

Hence, the correct answer is option D.

VIEW SOLUTION

40. **C**

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

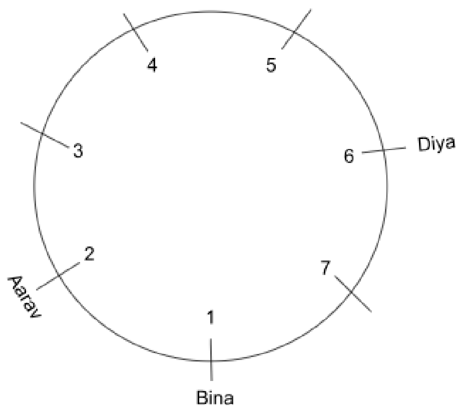
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

1) Passing the buck second to the right in round 4 and second to the right in round 5.
2) Passing the buck second to the left in round 4 and to the left in round 5.
3) Passing the buck to the left in round 4 and second to the left in round 5.

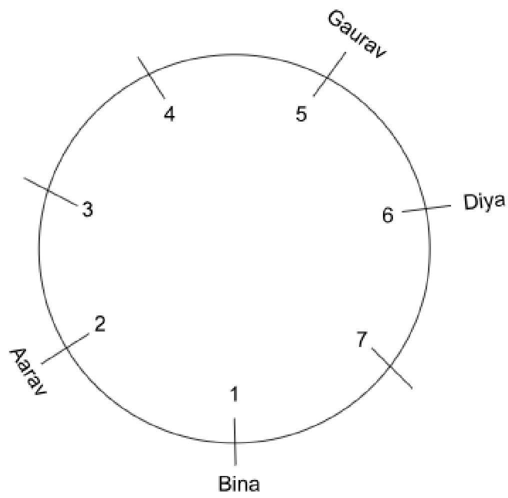
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.

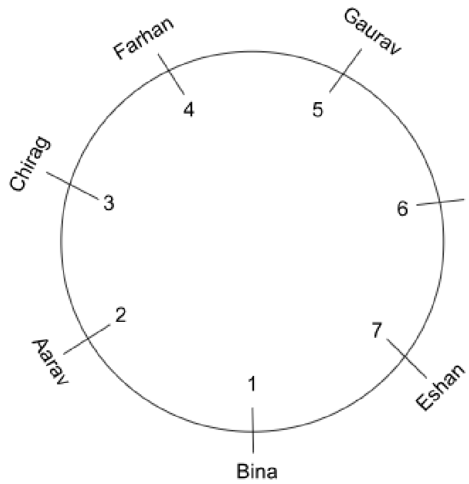


Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.
If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.
The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



This is the final arrangement.

The person third to the left of Eshan is Chirag.

Hence, the correct answer is option C.

VIEW SOLUTION

41. **C**

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

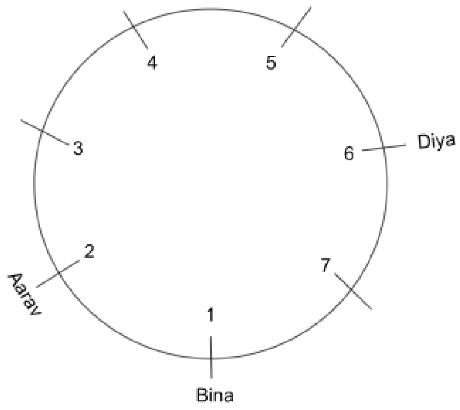
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

1) Passing the buck second to the right in round 4 and second to the right in round 5.
2) Passing the buck second to the left in round 4 and to the left in round 5.
3) Passing the buck to the left in round 4 and second to the left in round 5.

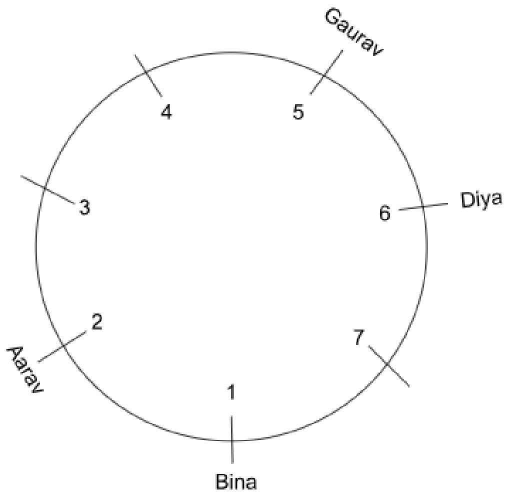
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.

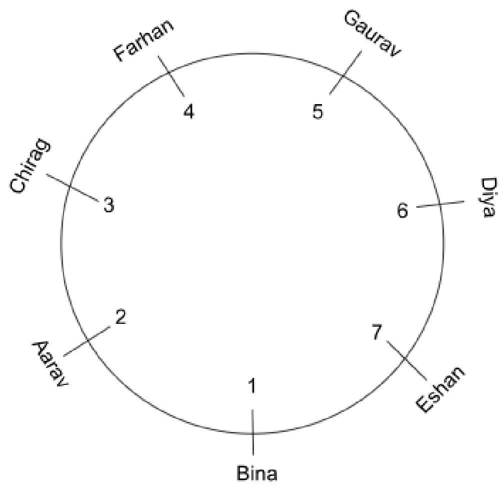


Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.
If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.
The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



In rounds 4 and 5, the possibilities were

1) Passing the buck second to the right in round 4 and second to the right in round 5.
2) Passing the buck second to the left in round 4 and to the left in round 5.
3) Passing the buck to the left in round 4 and second to the left in round 5.

Here, Buck could have been passed second to the right, second to the left or immediately to the left in rounds 4 and 5, which is unknown to us. The only pass type that we are certain to have happened is immediately to the right.

Hence, the correct answer is option C.

VIEW SOLUTION

42. **B**

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

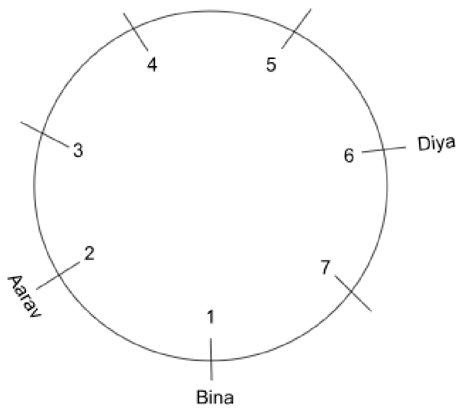
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

1) Passing the buck second to the right in round 4 and second to the right in round 5.
2) Passing the buck second to the left in round 4 and to the left in round 5.
3) Passing the buck to the left in round 4 and second to the left in round 5.

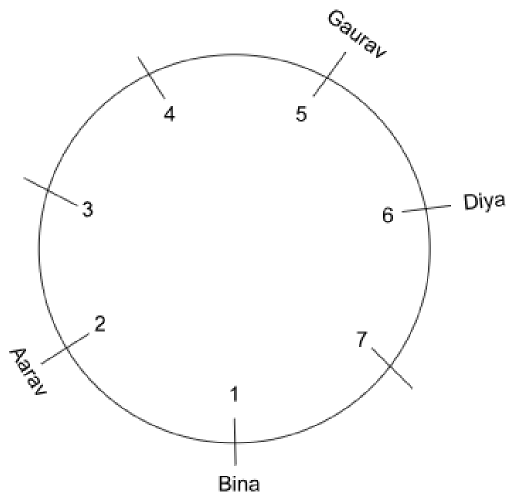
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.

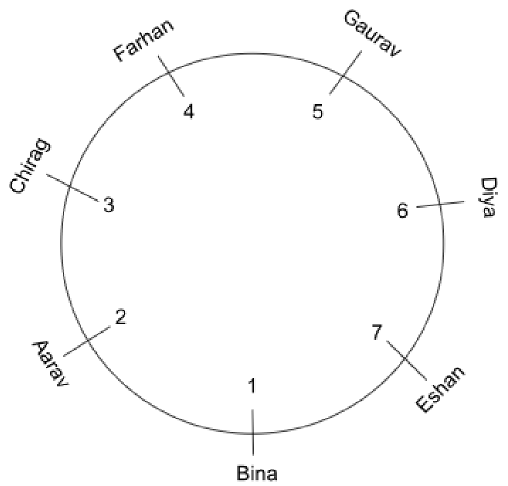


Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7.
If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.
The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7 and the only person left to be assigned is Eshan making the person at position 7 to be Eshan.



This is the final arrangement.

The person immediately to the right of Bina is Eshan.

Hence, the correct answer is option B.

VIEW SOLUTION

43. **D**
According to given condition we know that Alam started with 40 spent 35 and saved 5 , Daljeet started with 20 , ganesh started with 20 and saved 16.5 , Jugraj started with 10 and spent less than 3 rs , sandeep started with 30 and spent more than 1.5 rs. So among the options only option D satisfies given requirements.

VIEW SOLUTION

44. **C**

According to given conditions , following time table is possible:

SS	VA	SK	JKG	JKR	RAS
9-10	10-11	11-11:30	11:30-12:30	1:30-2:00	2:00

hence option C.

VIEW SOLUTION

45. **C**
We can get the following from the information provided.

	Student	City	School
1	Riaz	Hyderabad	Little Flower
2	Balbir	Bangalore	Loyola
3	Bipin	Pune	Convent

Hence, Option C is right.

VIEW SOLUTION

46. **C**

According to given conditions we have -

Sharmas	Banerjees	Pattabhiraman
12 noon	2:00 PM	1:00 PM
Sambar	makkey ki roti	fried brinjal
White chinaware	red chinaware	blue chinaware

VIEW SOLUTION

47. **B**
According to given question Jerome + Tommy = Albert + David eq.(1) ; Jerome < Tommy eq. (3)
Albert + Tommy < David + jerome eq. (2)
Now adding above two equations we will get
2David > 2Tommy
or David > Tommy
or Tommy > Jerome > Albert (from eq.(3))
Now David should earn most pocket money as to hold equation (2) to be true.
Hence David > Tommy > Jerome > Albert.

VIEW SOLUTION

48. **C**
It is given in the statements, that C cannot travel with A or B. E cannot travel with B or F. G cannot travel with F, or H, or D. By formulating the table we get

	A	B	C	D	E	F	G	H
A	×		×					
B		×	×		×			
C	×	×	×					
D				×			×	
E		×			×	×		
F					×	×	×	
G				×		×	×	×
H							×	×

Q is with A and G is with Q => G and Q are travelling together on motorcycle M3
F travels on M1 and E travels on M2 motorcycles.
D is travelling with P on M2 => D and E are traveling together on M2
B cannot go with R => F and B go together on M1
Therefore, C and H go together on M4

So, the table can formed as below :

	A	B	C	D	E	F	G	H
A	×	×	×	×	×	×	✓	×
B	×	×	×	×	×	✓	×	×
C	×	×	×	×	×	×	×	✓
D	×	×	×	×	✓	×	×	×
E	×	×	×	✓	×	×	×	×
F	×	✓	×	×	×	×	×	×
G	✓	×	×	×	×	×	×	×
H	×	×	✓	×	×	×	×	×

Basket	Motorcycle		
O	M1	F	B
P	M2	E	D
Q	M3	A	G
R	M4	C	H

Person	A	B	C	D	E	F	G	H
Basket	Q	O	R	P	P	O	Q	R
Motorcycle	M3	M1	M4	M2	M2	M1	M3	M4

Hence, C would be travelling with H.

VIDEO SOLUTION

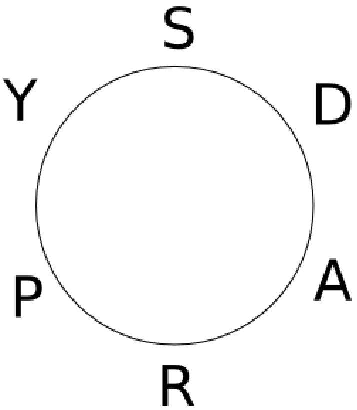
49. **C**

According to given conditions ,we get

Vaibhav	Suprita	Anshuman
June	April	September
7 yrs	4 yrs	2 yrs

VIEW SOLUTION

50. **C**
The correct original circular arrangement sequence in clockwise manner is Suresh, Dhiraj, Ajay, Raghuveer, Pramod, Yogendra.



So after the changes, Suresh is to the left of Dhiraj.

VIEW SOLUTION

51. **C**
From Statement I, we know that Ashish is not an engineer, and from Statement IV Dhanraj is not an engineer either. Since Sameer is an Economist, Felix must be the engineer.

Name	Ashish	Dhanraj	Felix	Sameer
Profession			Engineer	Economist

From Statement IV, Dhanraj got more offers than Ashish. Since Doctor got the most number of offers, Ashish is not the doctor:

Name	Ashish	Dhanraj	Felix	Sameer
Profession	CA	Doctor	Engineer	Economist

We know that the doctor got offers from 3 NIMs and Ashish got more offers than the engineer. Since Ashish is a CA and did not get 2 offers, he can only get 1 offer and the engineer gets none. The final table will look like this:

Name	Ashish	Dhanraj	Felix	Sameer
Profession	CA	Doctor	Engineer	Economist
No. of offers	1	3	0	2

Here, we can see that Sameer is an economist with offers from 2 NIMs. Hence only Option C is correct.

VIEW SOLUTION

52. **D**

Mrs. Abraham can't be sitting next to him as per the seating arrangement. Wives sit three places away from their husbands.

Mrs. Charlie is sitting to the left of Mr. Abraham. So, she can't be sitting to his right.

Mrs. Elenor is sitting two places to the right of Mrs. Border (and not Mrs. Charlie). So, she can't be sitting right next to Mr. Abraham.

Mrs. Border and Mrs. Dennis are the remaining two wives and each is equally likely to to be sitting to the right of Mr. Abraham.

VIDEO SOLUTION

53. **B**

Names	Language	Coherence	Subject-Matter	Machinary	Recent-develop.
Akhila		13			
Bhanu					
Charulata					
Divya				10	
Ela					

Given informations are as follows:

Akhila got 13 marks in coherence and divya got 10 marks in Machinary

Divya's marks = Ela's Marks (Hence they both are not winners as winner is only one)

Bhanu's marks < Akhila's marks (Hence bhanu is also not a winner)

So winner will be either Akhila or Charulata

As Akhila got 13 marks in coherence so she must have 20 marks in some head as highest marks are 90

That's why winner will be Charulata.

VIEW SOLUTION

54. **B**

According to given conditions, the groups are as follows:

Case 1:

W1 : Sonali, Shalini, Shubhra ,Shahira and Rupa
W4: Somya , Ruchika, Jyotika, sweta
: Renuka, Rupali, Komal
: Ritu and Tara
: Radha

Case 2:

W1 : Sonali, Shalini, Shubhra ,Shahira and Rupa
W4: Somya , Ruchika
: Renuka, Rupali, Komal
: Ritu, Tara, Jyotika, sweta
: Radha

Hence, Radha must be alone in a group . Therefore, her trainer is Elina.

VIEW SOLUTION

55. **D**

For min steps 1st move would be to place book 3 from A to B

2nd move would be to place book 2 from A to B,

3rd move would be to place book 2 and 3 from B to A.

Last move would be to place books 2,3,1 from table A to Table B.

So 4 moves are atleast needed.

VIEW SOLUTION

56. **B**

According to given condition we know that P likes Blue and red , M likes yellow , U likes Red and blue , T likes Black and x lives in a hotel . Since the person in palace doesnt like blue or black. Only 1 such person is possible i.e. M

VIEW SOLUTION

57. **A**

According to given condition the correct sequence of houses is 1st is blue, 2nd green, 3rd is yellow and last red.

Now in the yellow house i.e. 3rd Z lives and X doesnt live as a neighbour i.e. Green and Red house.

So, X lives in blue house.

VIEW SOLUTION

58. **B**

If A and B are together, then it can't be at first and second place as C and D will be together which is not possible.
Now if A can be at third place and B is at second place then C and D can be placed accordingly.
Answer will be B).

VIEW SOLUTION

59. **C**

If A is not at the third place, then it has only two choices either 2nd or 4th position
So if A is at 2nd position, then B has only fourth position as choice and C can have either first or third position.
And if A is at 4th position, then B has only 2nd position and again C can have either first or third position.
Hence answer will be C.

VIEW SOLUTION

60. **A**

Among all positions for A, if it has first position then either B will be at the 3rd position or C and D will be together which is not possible Hence answer will be A) .

VIEW SOLUTION

61. **C**

Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

VIEW SOLUTION

62. **B**

Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

VIEW SOLUTION

63. **A**

Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

VIEW SOLUTION

64. **C**

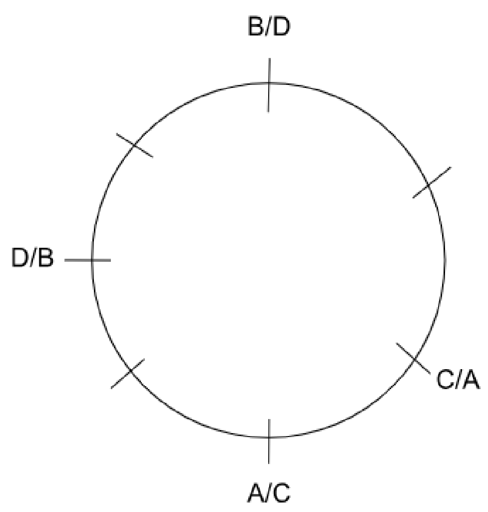
Since the people wearing yellow saree and the white saree were at the ends, Ms West Bengal was not sitting at one of the ends and was not a runner up, the arrangement is as shown below:

YELLOW (1)	RED	GREEN (2)	WHITE
ANDHRA PRADESH	WEST BENGAL	UTTAR PRADESH	MAHARASHTRA

VIEW SOLUTION

65. **A**

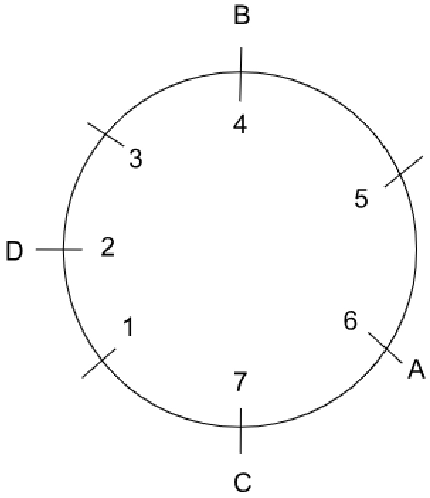
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

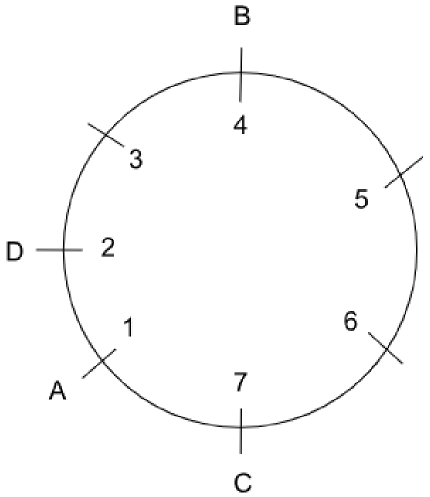
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

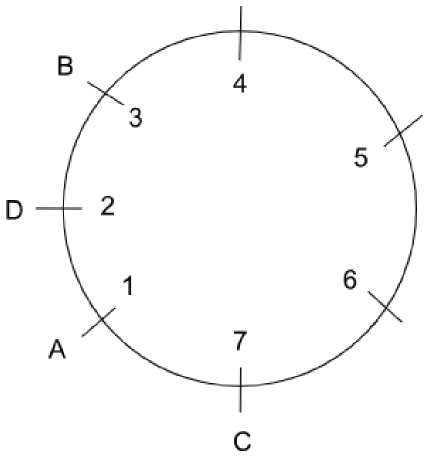
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



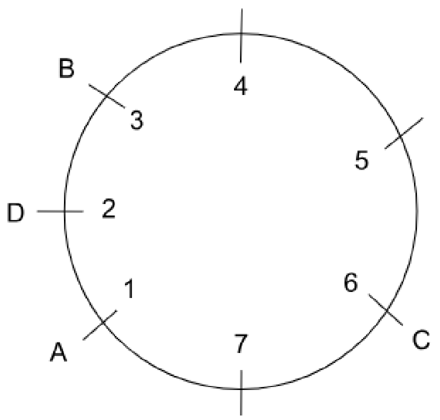
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



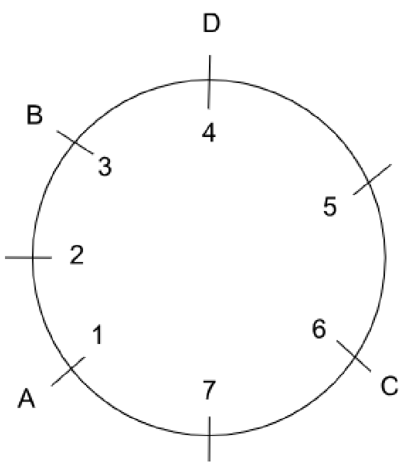
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

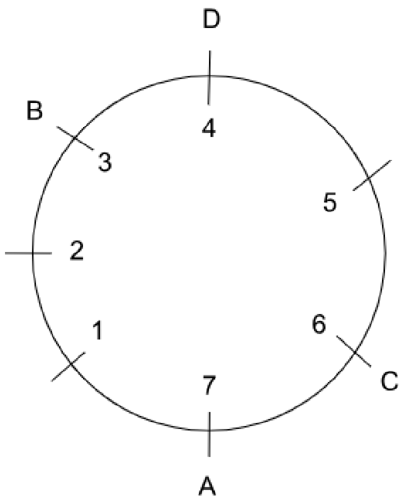
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

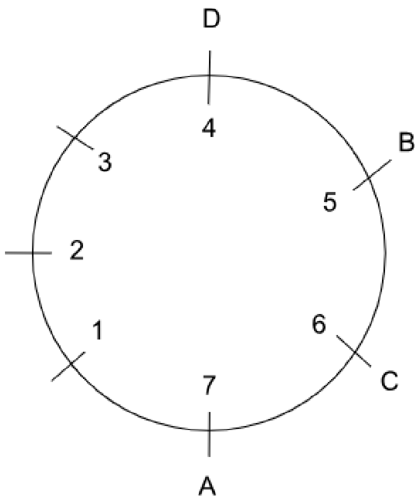
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



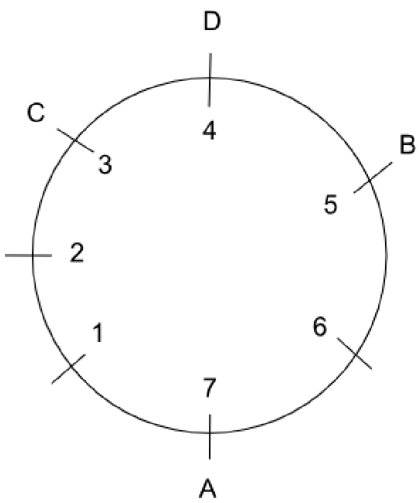
Turn 6: Bashir moves

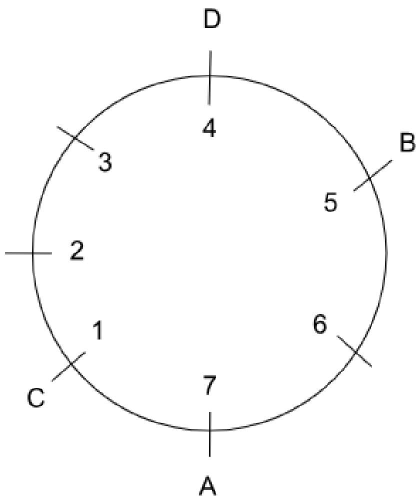
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





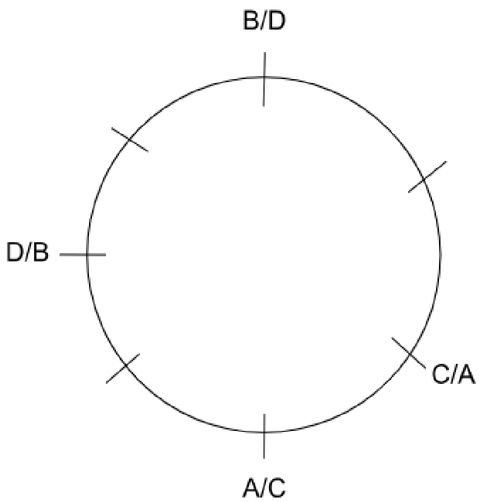
The chairs occupied after turn 6 are 4, 5, 6 and 7.

Hence, the correct answer is option A.

VIEW SOLUTION

66. **D**

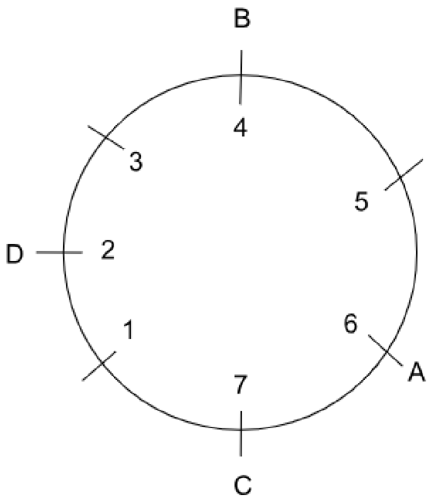
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

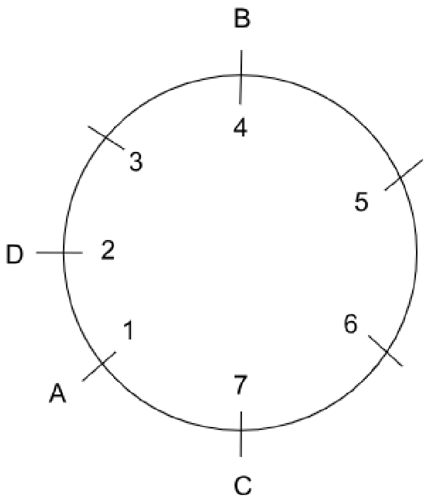
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

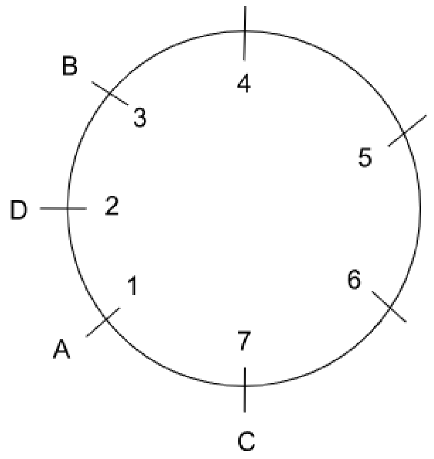
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



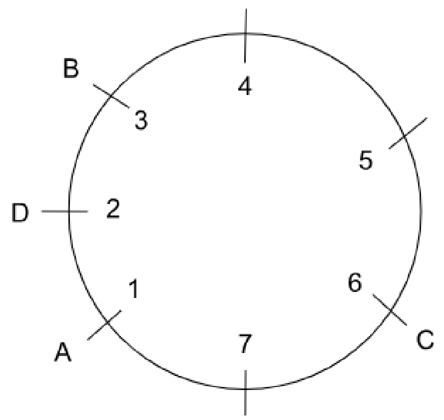
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



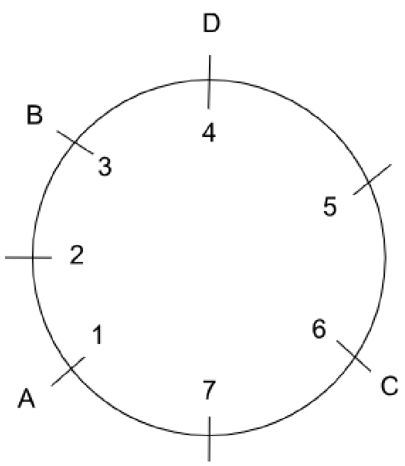
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

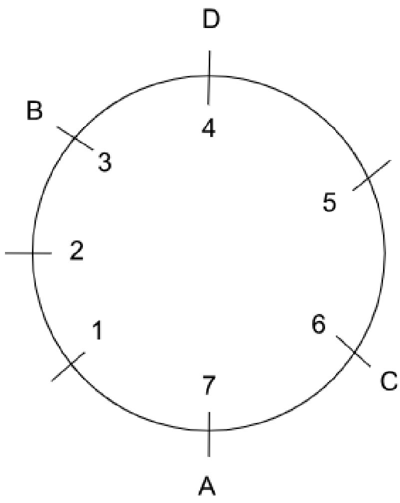
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

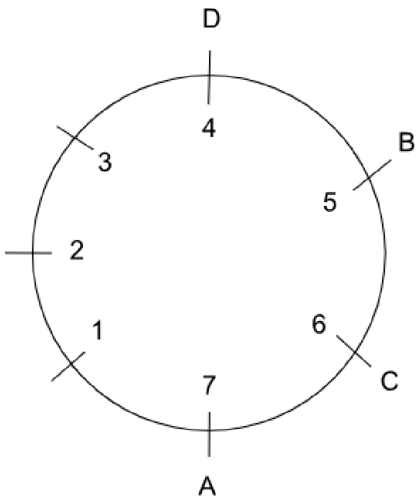
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



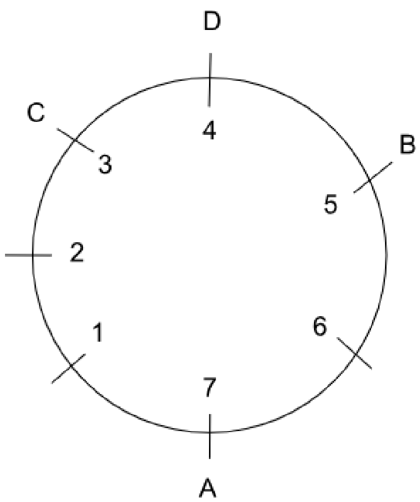
Turn 6: Bashir moves

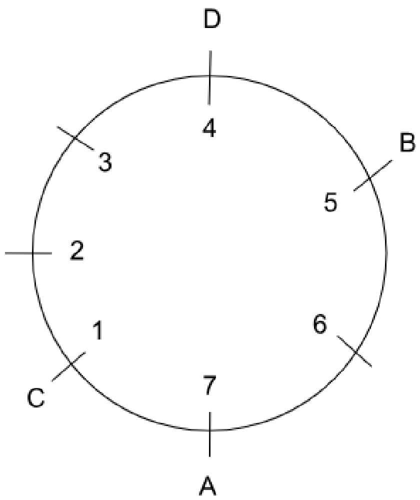
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





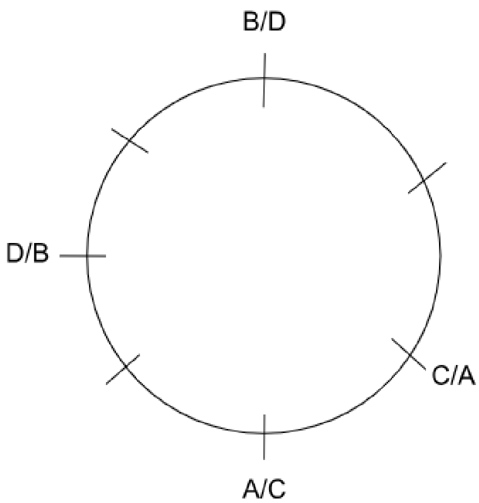
After turn 3, chair 4 is occupied by no one.

Hence, the correct answer is option D.

VIEW SOLUTION

67. **B**

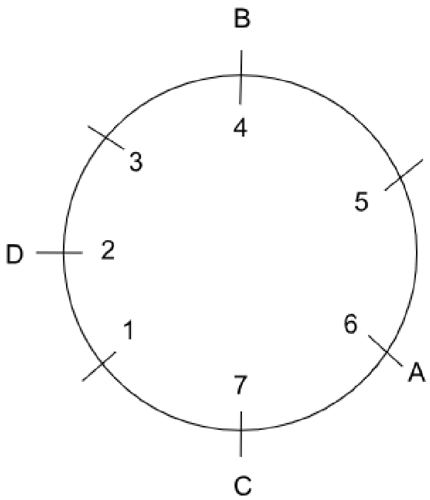
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



We now have the initial position, and we are told that all the members are adjacent only after turn 2 and turn 6.

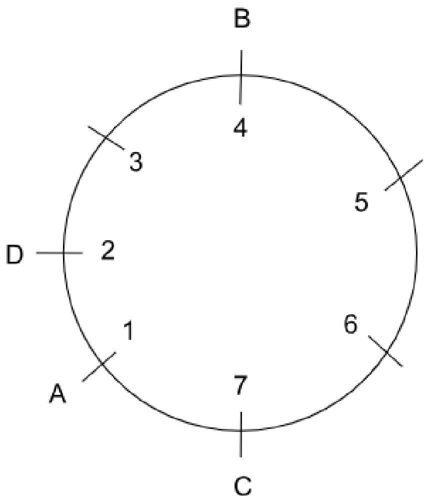
We know that A changed his position in turn 1.

Turn 1: Aslam moves

If A changed his position anticlockwise, he would go to chair 5, and if he changed his position clockwise, he would go to chair 1.

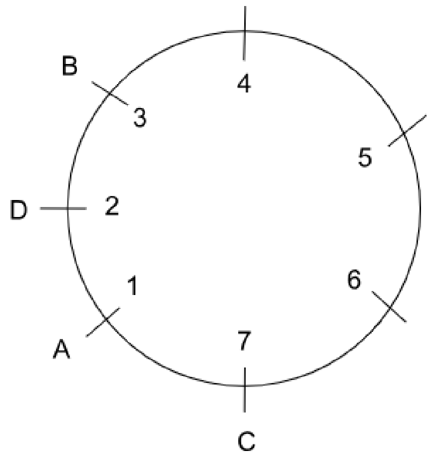
If A ends in chair 5 after turn 1, then it is not possible for all of them to be adjacent to each other after B changes his position in turn 2.

So, A has to end in chair 1 after turn 1, so that B will have a chance in turn 2 for all of them to be adjacent.



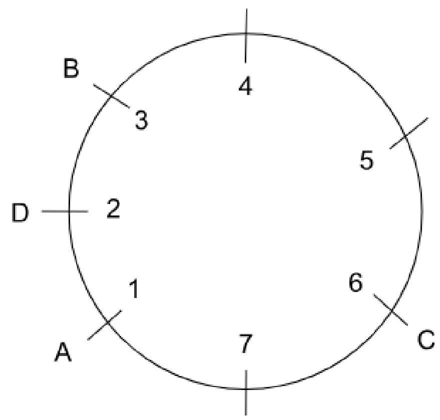
Turn 2: Bashir moves

For all of them to be adjacent after turn 2, B must change his position anticlockwise, and after turn 2, he would end up on chair 3.



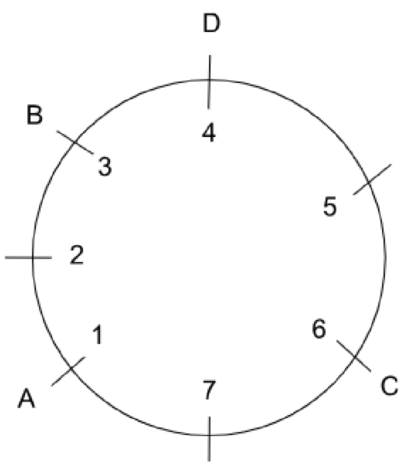
Turn 3: Chhavi moves

We are given that Davies occupies chair 4 after turn 5, and for this to happen, chair 4 must be empty and cannot be occupied by anyone who does not move before turn 5. So, in turn 3, C must go anticlockwise and occupy chair 6, as if he goes clockwise, he would occupy chair 4, which must be empty for Davies.



Turn 4: Davies moves

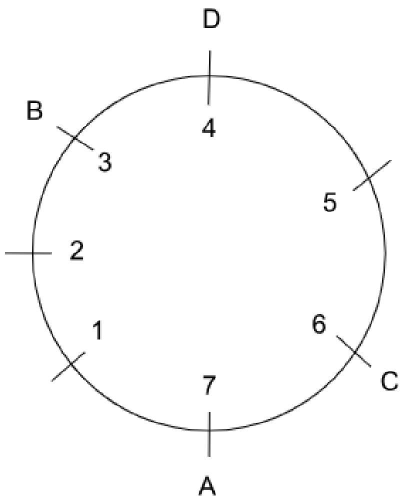
Davies will not move in turn 5, and for him to occupy chair 4 after turn 5, he must occupy it on turn 4. So, D has to go clockwise in turn 4 and must occupy chair 4.



Turn 5: Aslam moves

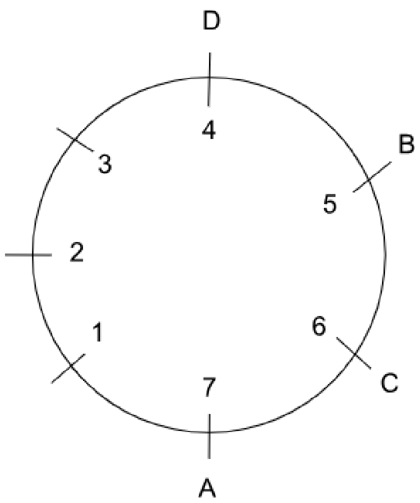
We know that all of them are again adjacent after turn 6. If A goes clockwise in turn 5, then he will occupy the chair 2, and in that case, it is not possible for them to be adjacent after turn 6 in either of the cases of B going clockwise or anticlockwise.

So, A must go anticlockwise and occupy chair 7 for them to be adjacent after turn 6.



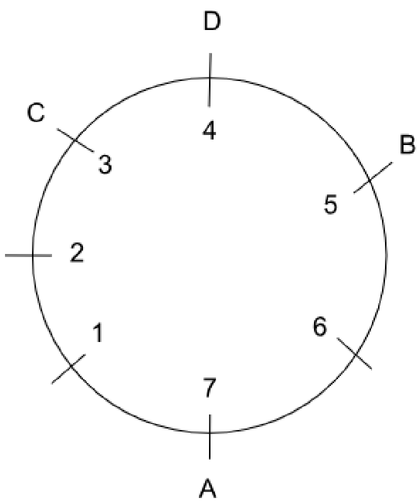
Turn 6: Bashir moves

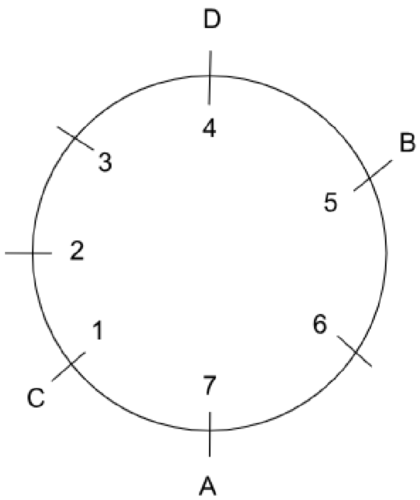
Now, for them to be adjacent after turn 6, B must go clockwise and must occupy chair 5.



Turn 7: Chavvi moves

We do not have any information to determine whether C went clockwise or anticlockwise. So, both cases are possible after turn 7.





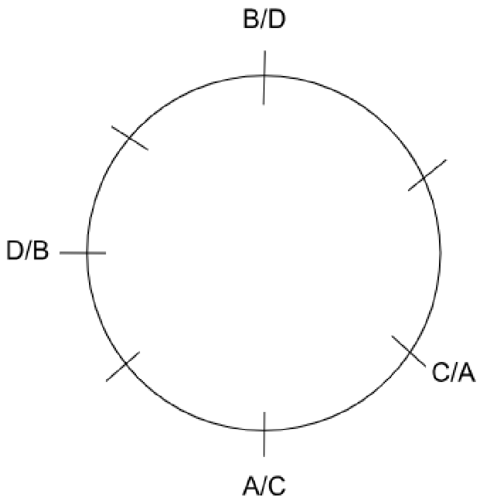
In either of the cases after turn 7, the seat adjacent to Bashir is occupied only by Davies, and the other one is empty.

Hence, the correct answer is option B.

VIEW SOLUTION

68. **4**

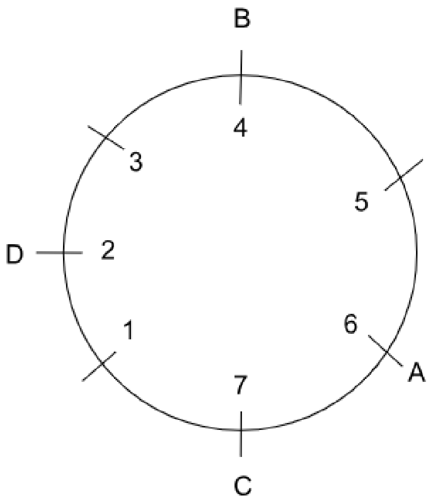
Based on the information about the initial positions of Aslam and Chhavi sitting next to each other, while Bashir and Davies have empty chairs on either side of their seats, the possible combinations are,



This has 4 possible combinations.

We are also given that Davies occupies Chair 2 after Turn 1 and Chhavi occupies Chair 7 after Turn 2. We know that the positions of Davies and Chhavi won't be changing after turn 2, as we know that in those turns, only the positions of Aslam and Bashir are being changed in the first 2 turns. Therefore, we can determine that Davies' initial position is chair 2, and Chhavi's initial position is chair 7.

Out of the 4 possible combinations, Davies on chair 2 and Chhavi on chair 7 are only possible in one case, which is,



So, Bashir's initial position is chair 4.

Hence, the correct answer is 4.

VIEW SOLUTION

69. **A**

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.
From (ii) No composer who assigned item to Rani, assigned any item to Samragni.

Hence Dyu will compose for Samragni 1st Performance and Gagan will compose for Queen 1st Performance. Also, Badal will compose for Samragni 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragni	Queen	Rani	Princess	Samragni	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

The sixth performance was composed by Badal. Hence C is the answer.

VIDEO SOLUTION

70. **D**

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.
From (ii) No composer who assigned item to Rani, assigned any item to Samragni.

Hence Dyu will compose for Samragni 1st Performance and Gagan will compose for Queen 1st Performance. Also, Badal will compose for Samragni 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragni	Queen	Rani	Princess	Samragni	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

The first and the sixth items were composed by Badal. Hence D is the answer.

VIDEO SOLUTION

71. **A**

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.
From (ii) No composer who assigned item to Rani, assigned any item to Samragni.

Hence Dyu will compose for Samragni 1st Performance and Gagan will compose for Queen 1st Performance. Also, Badal will compose for Samragni 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragni	Queen	Rani	Princess	Samragni	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

The second performance was composed by Dyu. Hence A is the answer.

VIDEO SOLUTION

72. **D**

	1st Performance				2nd Performance			
Finalists	Princess							Rani
Composers	Badal							Gagan

Since the dancers performed their second items in the same sequence of their performance of their first items. The table will be as follows:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal							Gagan

The items assigned by Ashman were performed consecutively. The number of performances between items assigned by each of the remaining composers was the same.

Also, the first items performed by the four dancers were all assigned by different music composers. Badal can come only at the place as shown in the table.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan			Badal		Gagan

Then Ashman can only compose for the following performances.

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal		Gagan	Ashman	Ashman	Badal		Gagan

Hence Dyu will compose for the following performances:

	1st Performance				2nd Performance			
Finalists	Princess			Rani	Princess			Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

From (i) No composer who assigned item to Princess, assigned any item to Queen.
From (ii) No composer who assigned item to Rani, assigned any item to Samragni.

Hence Dyu will compose for Samragni 1st Performance and Gagan will compose for Queen 1st Performance. Also, Badal will compose for Samragni 2nd Performance and Dyu will compose for Queens 2nd Performance.

Hence, the complete table is as follows:

	1st Performance				2nd Performance			
Finalists	Princess	Samragni	Queen	Rani	Princess	Samragni	Queen	Rani
Composers	Badal	Dyu	Gagan	Ashman	Ashman	Badal	Dyu	Gagan

Option A: Samragni did not perform in any item composed by Ashman. This statement is true.

Option B: Princess did not perform in any item composed by Dyu. This is also true.

Option C: Rani did not perform in any item composed by Badal. This statement is true.

Option D: Queen did not perform in any item composed by Gagan. This statement is false.

Hence D is the answer.

VIDEO SOLUTION

73. **D**

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

We can see that all options except option D are definitely true. Option D cannot be ascertained to be true. Dorm 10 can cost Rs. 1 crore or Rs. 6 crores to repair. Therefore, option D is the right answer.

VIDEO SOLUTION

74. **19**

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

Cost = 3 + 3 + 6 + 3 + 4 = Rs.19 crores. Therefore, 19 is the correct answer.

VIDEO SOLUTION

75. **3**

Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

There are 3 dorms from 6 to 10 which are women's dorms.

It has been given that the cost of repairing the woman dorms add up to 20. Therefore, the distribution of the costs should be 6+6+5+3.
Dorm 4 is the dorm whose number is below 5 but is a woman's dorm. Therefore, dorm 9 should cost Rs.3 crores to repair. Dorm 8 cannot be a woman's dorm. Therefore, dorm 10 should be a woman's dorm and should cost Rs. 6 crore to repair.

Dorm 9 will cost Rs.3 crore to repair and hence, 3 is the correct answer.

VIDEO SOLUTION

76. **D**
Odd numbered dorms need either moderate or extensive repair.

Even numbered dorms need either light or extensive repair.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 8, 10	1, 3, 5, 7, 9	1, 2, 4, 5, 7, 8, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)			

It has been given that dorms 4 to 9 all require different repairing costs. The dorms 3 and 9 should require moderate repair (going by the table). Dorm 7 costs the highest. Therefore, dorm 7 should require 6 crores to repair. Dorm 8 requires the least cost to repair. Therefore, dorm 8 should cost 1 crore to repair. We can eliminate these dorm numbers from other 2 lists.

Type of repair	Light	Moderate	Extensive
(Possible dorms)	2, 4, 6, 10		2, 4, 10
Distribution	2 (1 crore) 1 (2 crore)	3 (3 crore) 1 (4 crore)	1 (5 crore) 2 (6 crore)
Dorm numbers (Definite)	8	1, 3, 5, 9	7

Dorms 4 to 9 cost different costs to repair. => Both dorms 5 and 9 cannot require the same cost of repair. Dorms 1 and 3 should require 3 crores to repair.

Dorm 6 should require light repair (2 crores) since dorm 8 requires 1 crore to repair.
=> Dorm 4 requires 5 crore to repair.

Dorm	1	2	3	4	5	6	7	8	9	10
Cost	3	1 or 6	3	5	3 or 4	2	6	1	3 or 4	1 or 6

It has been given that the cost of repairing the woman dorms add up to 20. Therefore, the distribution of the costs should be 6+6+5+3.
Dorm 4 is the dorm whose number is below 5 but is a woman's dorm. Therefore, dorm 9 should cost Rs.3 crores to repair. Dorm 8 cannot be a woman's dorm. Therefore, dorm 10 should be a woman's dorm and should cost Rs. 6 crore to repair.

Hence, Option D is the right answer.

VIDEO SOLUTION

77. **7**
Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.
Let's take cases:-
1. The lowest rating is 4
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is not possible as there are only 3 odd numbers from 5-10.
Thus, the lowest rating is not 4.
2. The lowest rating is 2.
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is possible when the odd ratings are 3,5,7 and 9.
We are given that the highest rating is not even. Thus, 10 rating is not possible.
We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.
Thus, the rating of the tea in cup 3 is an even number.
Thus, the rating of the tea in cup 5 must be an odd number.
Only 1 such pair is possible of 3 and 6.
Thus, the tea in cup 2 got the rating of 2.
The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.
We are given that:-
Tea in Cup 3 got a higher rating than that in Cup 1.
Thus, the tea in cup 1 got a rating of 5.

Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.
Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.
The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

Hence, 7 is the 2nd highest rating given.

VIDEO SOLUTION

78. **4**

Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.
Let's take cases:-
1. The lowest rating is 4
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is not possible as there are only 3 odd numbers from 5-10.
Thus, the lowest rating is not 4.
2. The lowest rating is 2.
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is possible when the odd ratings are 3,5,7 and 9.
We are given that the highest rating is not even. Thus, 10 rating is not possible.
We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.
Thus, the rating of the tea in cup 3 is an even number.
Thus, the rating of the tea in cup 5 must be an odd number.
Only 1 such pair is possible of 3 and 6.
Thus, the tea in cup 2 got the rating of 2.
The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.
We are given that:-
Tea in Cup 3 got a higher rating than that in Cup 1.
Thus, the tea in cup 1 got a rating of 5.
Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.
Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.
The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

Thus, 4 was the number of the cup that contained tea from Ooty.

VIDEO SOLUTION

79. **B**

Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.
Let's take cases:-
1. The lowest rating is 4
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is not possible as there are only 3 odd numbers from 5-10.
Thus, the lowest rating is not 4.
2. The lowest rating is 2.
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is possible when the odd ratings are 3,5,7 and 9.

We are given that the highest rating is not even. Thus, 10 rating is not possible.
We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.
Thus, the rating of the tea in cup 3 is an even number.
Thus, the rating of the tea in cup 5 must be an odd number.
Only 1 such pair is possible of 3 and 6.
Thus, the tea in cup 2 got the rating of 2.
The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.
We are given that:-
Tea in Cup 3 got a higher rating than that in Cup 1.
Thus, the tea in cup 1 got a rating of 5.
Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.
Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.
The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

If the tea from Munnar did not get the minimum rating then it must have got the 2nd lowest rating as we know, Assam>Wyanand>Munnar.
Thus, Wyanand must have got a rating of 5.

VIDEO SOLUTION

80. **B**
Now we are given that the lowest rating is an even number and only 2 cups got an even number rating.
Let's take cases:-
1. The lowest rating is 4
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is not possible as there are only 3 odd numbers from 5-10.
Thus, the lowest rating is not 4.
2. The lowest rating is 2.
If the lowest rating in 4 then the other ratings will be in the range 5-10.
From this we need 4 odd and 1 even numbers.
This is possible when the odd ratings are 3,5,7 and 9.
We are given that the highest rating is not even. Thus, 10 rating is not possible.
We are also given that the rating of tea in Cup 3 was double the rating of the tea in Cup 5.
Thus, the rating of the tea in cup 3 is an even number.
Thus, the rating of the tea in cup 5 must be an odd number.
Only 1 such pair is possible of 3 and 6.
Thus, the tea in cup 2 got the rating of 2.
The tea in cup 3 got a rating of 6 and the tea in cup 5 got a rating of 3.
We are given that:-
Tea in Cup 3 got a higher rating than that in Cup 1.
Thus, the tea in cup 1 got a rating of 5.
Cup 6 contained tea from Himachal and the Tea from Ooty got the highest rating.
Thus, cup 6 got a rating of 7 and cup 4 got a rating of 9.
The table is as shown below:-

Cup no.	1	2	3	4	5	6
Rating	5	2	6	9	3	7
Place				Ooty		Himachal

It is given that the rating of Assam>Wayanad>Munnar

Hence, since Wayanad and Ooty are in consecutive cups, Wayanad can be either in cup number 3 or 5.

So Wayanad can only be in cup number 5, then Munnar will be in cup number 2. So Darjeeling and Assam can be in cup 1 and 3 in any order.

Hence B is a possibility.

VIDEO SOLUTION

81. **A**
We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.
Let us see how the friends are supposed to pay for the seats they choose:-
In row 1-1000
In row 2-10 - 300 for window and 500 for aisle
In row 11 - 200 for window and 400 for aisle
In row 12,13 - 1000
In row 14-20 - 200 for window and 400 for aisle
In row 21-30 - 0
Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.
Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.
Manik sat beside Jayantha and thus Manik is also sitting in row 10.
Now we are given that 7 of the 8 friends paid a total of 4600 Rs.
Let's start with the cases:-
It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600
Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.
This is possible only when each of them pay 200 Rs.
So the case is- 1000*4 , 200*3
Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.
There are 2 cases where this is possible:-
1000*3, 500*2, 400, 200
1000*3, 400*4
Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.
This is not possible as each friend can pay a maximum of 500 Rs.
Thus, the possible cases are
1000*4 , 200*3
1000*3, 500*2, 400, 200
1000*3, 400*4
As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.
Thus, Jayantha paid 500 Rs.
Thus, the case is:-
1000*3, 500*2, 400, 200
Thus, Manik must have also paid 500 sitting in row 10 aisle seat
Ajit must be sitting in row 11 aisle seat paying 400 Rs.
Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.
Thus, among Gargi, Kikira, Pradosh and Tapesh 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.
Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.
Among Pradosh and Tapesh a maximum of 1 person could have paid 1000 Rs.
Thus, the only possible case here is :-
Gargi and Kikira paid 1000 each.
Pradosh is sitting ahead of Tapesh and one of them paid 200 Rs.

Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapesh sitting in row 21 paying nothing.
Thus, the amount paid by each friend is as shown below:

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapesh	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Manik is sitting in row 10.

VIDEO SOLUTION

82. **D**
We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.
Let us see how the friends are supposed to pay for the seats they choose:-
In row 1-1000
In row 2-10 - 300 for window and 500 for aisle
In row 11 - 200 for window and 400 for aisle
In row 12,13 - 1000
In row 14-20 - 200 for window and 400 for aisle
In row 21-30 - 0
Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.
Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.
Manik sat beside Jayantha and thus Manik is also sitting in row 10.
Now we are given that 7 of the 8 friends paid a total of 4600 Rs.
Let's start with the cases:-
It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600
Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.
This is possible only when each of them pay 200 Rs.
So the case is- 1000*4 , 200*3
Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.
There are 2 cases where this is possible:-
1000*3, 500*2, 400, 200
1000*3, 400*4
Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.
This is not possible as each friend can pay a maximum of 500 Rs.
Thus, the possible cases are
1000*4 , 200*3
1000*3, 500*2, 400, 200
1000*3, 400*4
As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.
Thus, Jayantha paid 500 Rs.
Thus, the case is:-

1000*3, 500*2, 400, 200
Thus, Manik must have also paid 500 sitting in row 10 aisle seat
Ajit must be sitting in row 11 aisle seat paying 400 Rs.
Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.
Thus, among Gargi, Kikira, Pradosh and Tapesh 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.
Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.
Among Pradosh and Tapesh a maximum of 1 person could have paid 1000 Rs.
Thus, the only possible case here is :-
Gargi and Kikira paid 1000 each.
Pradosh is sitting ahead of Tapesh and one of them paid 200 Rs.
Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapesh sitting in row 21 paying nothing.
Thus, the amount paid by each friend is as shown below:-

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapesh	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Gargi paid 1000 rs for her choice of seat

VIDEO SOLUTION

83. **C**
We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.
Let us see how the friends are supposed to pay for the seats they choose:-
In row 1-1000
In row 2-10 - 300 for window and 500 for aisle
In row 11 - 200 for window and 400 for aisle
In row 12,13 - 1000
In row 14-20 - 200 for window and 400 for aisle
In row 21-30 - 0
Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.
Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.
Manik sat beside Jayantha and thus Manik is also sitting in row 10.
Now we are given that 7 of the 8 friends paid a total of 4600 Rs.
Let's start with the cases:-
It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600
Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.
This is possible only when each of them pay 200 Rs.
So the case is- 1000*4 , 200*3
Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.

There are 2 cases where this is possible:-
1000*3, 500*2, 400, 200
1000*3, 400*4
Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.
This is not possible as each friend can pay a maximum of 500 Rs.
Thus, the possible cases are
1000*4 , 200*3
1000*3, 500*2, 400, 200
1000*3, 400*4
As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.
Thus, Jayantha paid 500 Rs.
Thus, the case is:-
1000*3, 500*2, 400, 200
Thus, Manik must have also paid 500 sitting in row 10 aisle seat
Ajit must be sitting in row 11 aisle seat paying 400 Rs.
Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.
Thus, among Gargi, Kikira, Pradosh and Tapesh 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.
Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.
Among Pradosh and Tapesh a maximum of 1 person could have paid 1000 Rs.
Thus, the only possible case here is :-
Gargi and Kikira paid 1000 each.
Pradosh is sitting ahead of Tapesh and one of them paid 200 Rs.
Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapesh sitting in row 21 paying nothing.
Thus, the amount paid by each friend is as shown below:-

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapesh	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Jayanta paid 500 for her choice of seat.

VIDEO SOLUTION

84. **D**
We are given that Jayanta, Ajit and Byomkesh were sitting in seats marked by the same letter, in consecutive rows in increasing order of row numbers; but all of them paid different amounts for their choices of seat.

Let us see how the friends are supposed to pay for the seats they choose:-
In row 1-1000
In row 2-10 - 300 for window and 500 for aisle
In row 11 - 200 for window and 400 for aisle

In row 12,13 - 1000
In row 14-20 - 200 for window and 400 for aisle
In row 21-30 - 0
Thus, As we can see 10, 11 and 12 are the only consecutive seats in which the amounts is different.
Thus, Jayanth, Ajit and Byomkesh sat in row 10, row 11 and row 12.
Manik sat beside Jayantha and thus Manik is also sitting in row 10.
Now we are given that 7 of the 8 friends paid a total of 4600 Rs.
Let's start with the cases:-
It is obvious that 5 friends cannot pay 1000 Rs for their seat because the amount will exceed 4600
Case 1:- 4 friends pay 1000 Rs each. Thus, the remaining friends will pay 600 Rs.
This is possible only when each of them pay 200 Rs.
So the case is- 1000*4 , 200*3
Case 2 :- 3 friends pay 1000 Rs each. Thus, the remaining friends will pay 1600 Rs.
There are 2 cases where this is possible:-
1000*3, 500*2, 400, 200
1000*3, 400*4
Case 3:- 2 friends pay 1000 Rs each. Thus, the remaining 5 friends will pay 2600 Rs.
This is not possible as each friend can pay a maximum of 500 Rs.
Thus, the possible cases are
1000*4 , 200*3
1000*3, 500*2, 400, 200
1000*3, 400*4
As there is no case in which a friend has to pay 300 Rs thus, Jayantha must be sitting in row 10 aisle seat.
Thus, Jayantha paid 500 Rs.
Thus, the case is:-
1000*3, 500*2, 400, 200
Thus, Manik must have also paid 500 sitting in row 10 aisle seat
Ajit must be sitting in row 11 aisle seat paying 400 Rs.
Byomyesh must be sitting row 12 aisle seat paying 1000 Rs.
Thus, among Gargi, Kikira, Pradosh and Tapesh 2 must have paid 1000, 1 must have paid 200 and the remaining person must have paid nothing.
Now we know Gargi and Kikira are sitting adjacent to each other and thus, either both or none of them must have paid 1000 Rs.
Among Pradosh and Tapesh a maximum of 1 person could have paid 1000 Rs.
Thus, the only possible case here is :-
Gargi and Kikira paid 1000 each.
Pradosh is sitting ahead of Tapesh and one of them paid 200 Rs.
Since, both of them were sitting in seats marked by the same letter, in consecutive rows thus, the only possibility is Pradosh sitting in row 20 window seat and paying 200 and Tapesh sitting in row 21 paying nothing.
Thus, the amount paid by each friend is as shown below:-

	Amount paid	Row-seat
Gargi	1000	1 or 13
Kikira	1000	1 or 13
Pradosh	200	20-W
Tapesh	0	21
Manik	500	10-A
Jayantha	500	10-A
Ajit	400	11-A
Byomyesh	1000	12-A

Tapesh did not paid any amount.

VIDEO SOLUTION

85. **C**

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let '4x' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that 'y' is the number of educationalists in the research committee and 'd' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let '5z' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

$\Rightarrow$ 10x+3y+5z = 24

We can see that each of x, y and z has to a natural number integer. If x > 1, then both y and z can't take any natural number.

Hence, we can say that x = 1.

At x = 1, 3y+5z = 14. If y = 1 or 2, Z is not an integer.

At x = 1 and y = 3, z = 1 which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

We can see that 'd' can assume two possible values. d = 1 or 2.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

Let us check the option one by one.

Option A: In the teaching committee the number of educationalists is equal to the number of politicians. We can see that in the teaching committee the number of educationalists can be equal to the number of politicians when both the numbers are '1'. Hence, this statement can be correct.

Option B: In the administration committee the number of bureaucrats is equal to the number of educationalists. We can see that in the administration committee the number of bureaucrats can be equal to the number of educationalists when both the numbers are '4'. Hence, this statement can be correct.

Option C: The size of the research committee is less than the size of the teaching committee. We can see the maximum size of teaching committee can be '6' which is less than the size of the research committee. Hence, the sentence is incorrect.

Option D: The size of the research committee is less than the size of the administration committee. We can see the minimum size of Administration committee can be '11' which is more than the size of the research committee. Hence, this statement is correct.

Therefore, we can say that option C is the correct answer.

VIDEO SOLUTION

86. **4**

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let '4x' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that 'y' is the number of educationalists in the research committee and 'd' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let '5z' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

$\Rightarrow$ 10x+3y+5z = 24

We can see that each of x, y and z has to a natural number integer. If x > 1, then both y and z can't take any natural number.

Hence, we can say that x = 1.

At x = 1, 3y+5z = 14. If y = 1 or 2, Z is not an integer.

At x = 1 and y = 3, z = 1 which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

We can see that 'd' can assume two possible values. d = 1 or 2.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

From the table, we can see that the number of bureaucrats in the administration committee = 4.

VIDEO SOLUTION

87. **3**

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let '4x' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that 'y' is the number of educationalists in the research committee and 'd' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let '5z' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

⇒ 10x+3y+5z = 24

We can see that each of x, y and z has to a natural number integer. If x > 1, then both y and z can't take any natural number.

Hence, we can say that x = 1.

At x = 1, 3y+5z = 14. If y = 1 or 2, Z is not an integer.

At x = 1 and y = 3, z = 1 which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

We can see that 'd' can assume two possible values. d = 1 or 2.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

From the table, we can see that the number of educationalists in the research committee = 3.

VIDEO SOLUTION

88. **A**

Let us draw a table according to the information given.

	Research	Teaching	Administration	Total
Bureaucrats				
Educationalists				
Politicians				
Total				24

It is given that the numbers of bureaucrats in the research and teaching committees are equal, while the number of bureaucrats in the research committee is 75% of the number of bureaucrats in the administration committee. Let '4x' be the number of bureaucrats in Administration committee.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists				
Politicians				
Total				24

The number of educationalists in the teaching committee is less than the number of educationalists in the research committee. The number of educationalists in the research committee is the average of the numbers of educationalists in the other two committees. Let us assume that 'y' is the number of educationalists in the research committee and 'd' be the difference in the number of educationalists in Research and teaching committees.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians				
Total				24

60% of the politicians are in the administration committee, and 20% are in the teaching committee. Let '5z' be the number of total number of politicians.

	Research	Teaching	Administration	Total
Bureaucrats	3x	3x	4x	10x
Educationalists	y	y-d	y+d	3y
Politicians	z	z	3z	5z
Total				24

We can say that

⇒ 10x+3y+5z = 24

We can see that each of x, y and z has to a natural number integer. If x > 1, then both y and z can't take any natural number.

Hence, we can say that x = 1.

At x = 1, 3y+5z = 14. If y = 1 or 2, Z is not an integer.

At x = 1 and y = 3, z = 1 which is the only possible solution.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	3-d	3+d	9
Politicians	1	1	3	5
Total				24

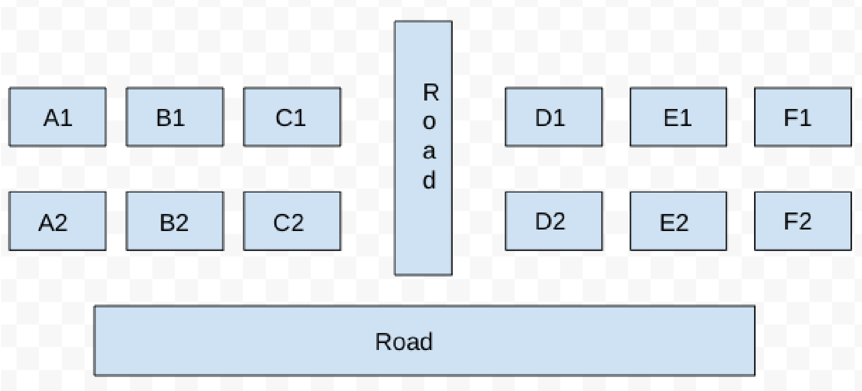
We can see that 'd' can assume two possible values. d = 1 or 2.

	Research	Teaching	Administration	Total
Bureaucrats	3	3	4	10
Educationalists	3	2/1	4/5	9
Politicians	1	1	3	5
Total	7	6/5	11/12	24

From the table, we can not uniquely determine the size of the teaching committee. Hence, option A is the correct answer.

VIDEO SOLUTION

89. **C**



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

=> 12+5a+3b = 24 => 5a+3b = 12 (a = road adjacency value, b= neighbor count)

The only value for which the equation satisfies is (a = 0, and b=4). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

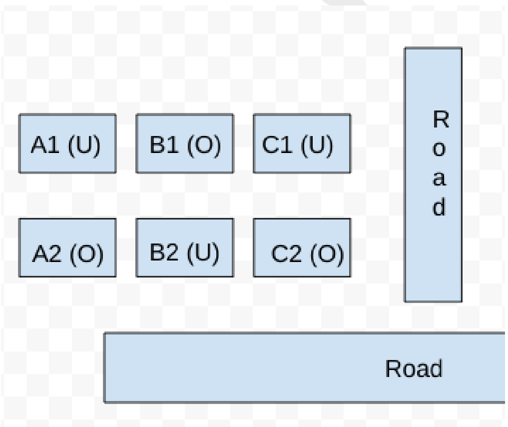
=> 10+5a+3b = 24 => 5a+3b = 14

=> (a, b) = (1, 3)

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and U = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: 12+5*1+3*b = 15 => 3b = -2 (which is not possible)

Hence, the house has no parking space => 10+5*1+3b = 15 => b = 0

b = 0 implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

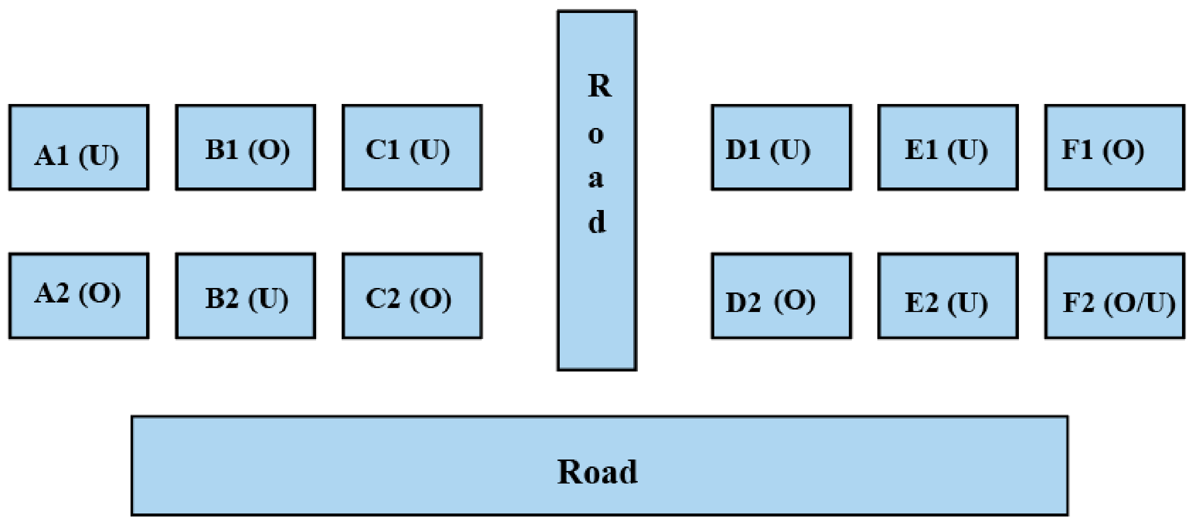
i) If the house has no working space, then: 10+5*0+3b = 15 => b = 5/3 (this is not possible since b has to be an integer value)

Hence, the house has parking space => 12+5*0+3b = 15 => b = 1 => One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10(there is no parking space)+ (5*0) + (3*the number of neighbours)
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.
Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.
Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:

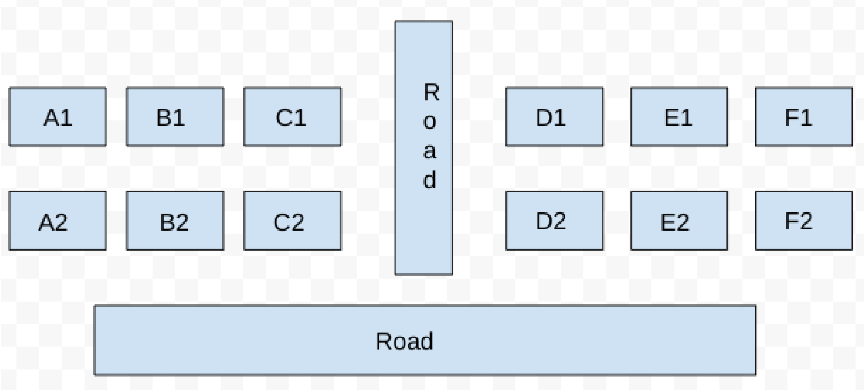


From the diagram, we can see that B1 and D2 are definitely occupied. The rest of the options are not definitely correct.

The correct options are both B and C.

VIDEO SOLUTION

90. **3**



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

=> 12+5a+3b = 24 => 5a+3b = 12 (a = road adjacency value, b= neighbor count)

The only value for which the equation satisfies is (a = 0, and b=4). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

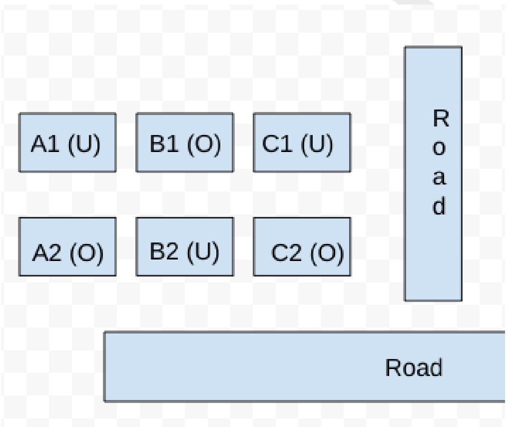
=> 10+5a+3b = 24 => 5a+3b = 14

=> (a, b) = (1, 3)

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and U = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: 12+5*1+3*b = 15 => 3b = -2 (which is not possible)

Hence, the house has no parking space => 10+5*1+3b = 15 => b = 0

b = 0 implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

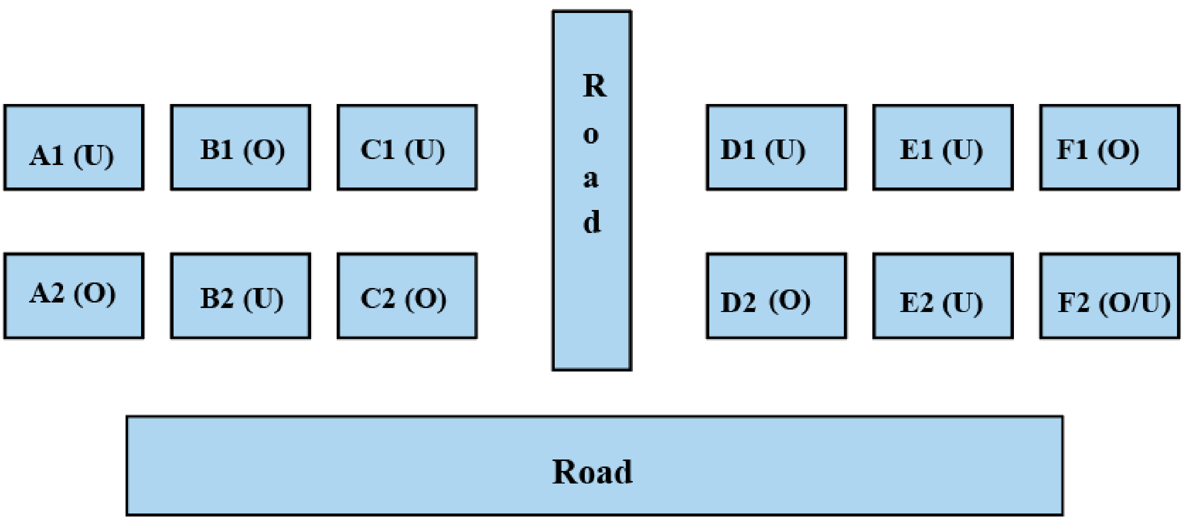
i) If the house has no working space, then: 10+5*0+3b = 15 => b = 5/3 (this is not possible since b has to be an integer value)

Hence, the house has parking space => 12+5*0+3b = 15 => b = 1 => One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10(there is no parking space)+ (5*0) + (3*the number of neighbours)
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.
Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.
Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

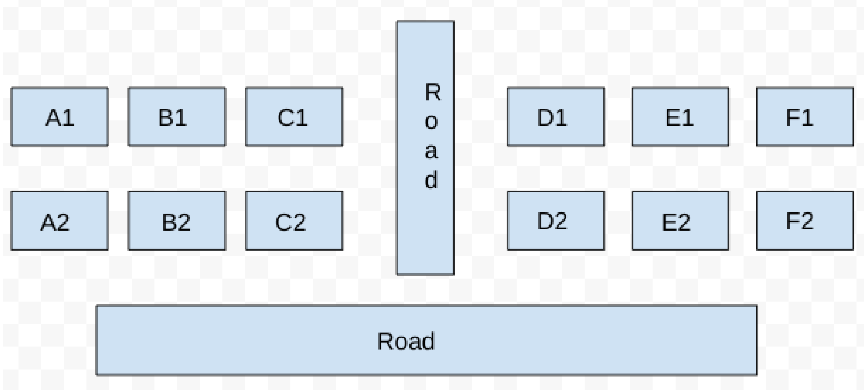
Therefore, the final diagram is given below:



From the diagram, we can see that 3 houses are vacant in block XX.

VIDEO SOLUTION

91. **D**



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

=> 12+5a+3b = 24 => 5a+3b = 12 (a = road adjacency value, b= neighbor count)

The only value for which the equation satisfies is (a = 0, and b=4). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

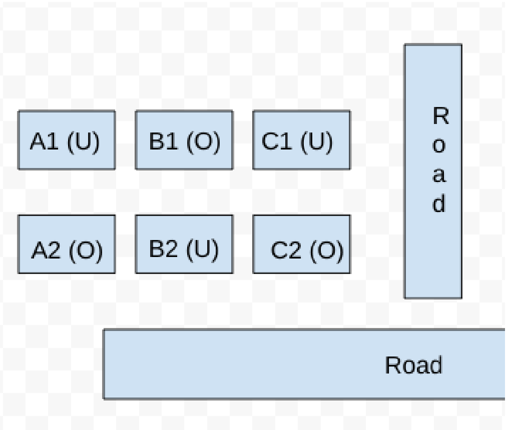
=> 10+5a+3b = 24 => 5a+3b = 14

=> (a, b) = (1, 3)

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and U = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: 12+5*1+3*b = 15 => 3b = -2 (which is not possible)

Hence, the house has no parking space => 10+5*1+3b = 15 => b = 0

b = 0 implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

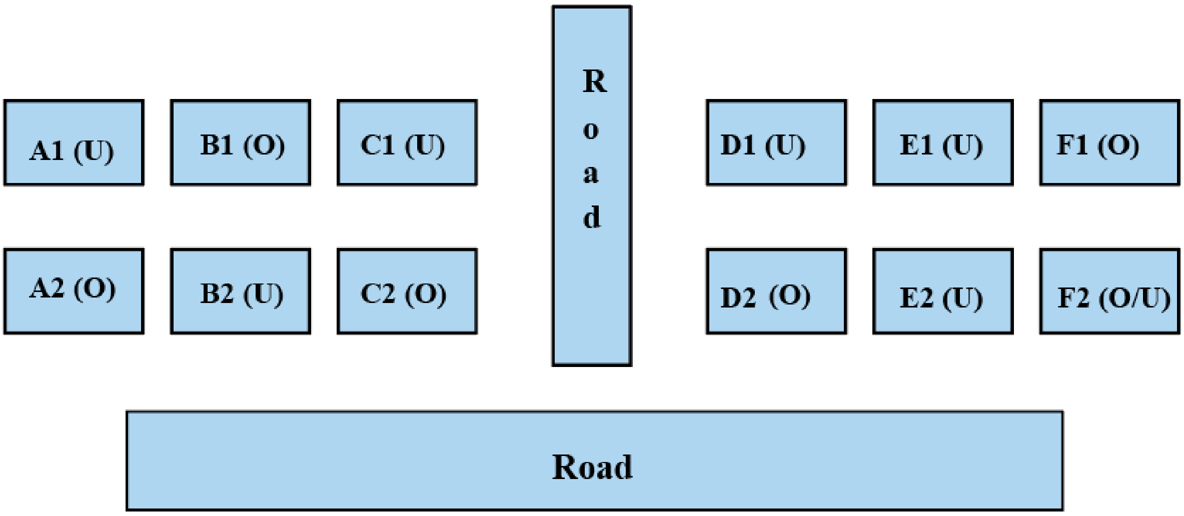
i) If the house has no working space, then: 10+5*0+3b = 15 => b = 5/3 (this is not possible since b has to be an integer value)

Hence, the house has parking space => 12+5*0+3b = 15 => b = 1 => One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10(there is no parking space)+ (5*0) + (3*the number of neighbours)
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.
Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.
Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:



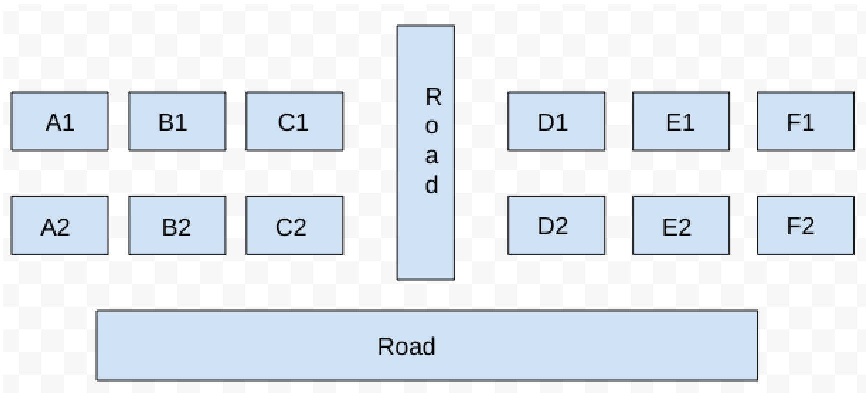
From the diagram, we can say that the number of vacant houses in Row 2 in Block XX is 1, and the number of vacant houses in Row 2 in Block YY is either 1 or 2.

Hence, the total number of vacant houses is either 2 or 3

The correct option is D

VIDEO SOLUTION

92. **21**



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

=> 12+5a+3b = 24 => 5a+3b = 12 (a = road adjacency value, b= neighbor count)

The only value for which the equation satisfies is (a = 0, and b=4). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

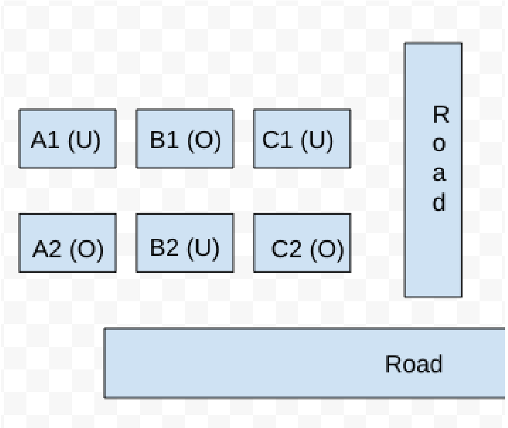
=> 10+5a+3b = 24 => 5a+3b = 14

=> (a, b) = (1, 3)

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and U = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: 12+5*1+3*b = 15 => 3b = -2 (which is not possible)

Hence, the house has no parking space => 10+5*1+3b = 15 => b = 0

b = 0 implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

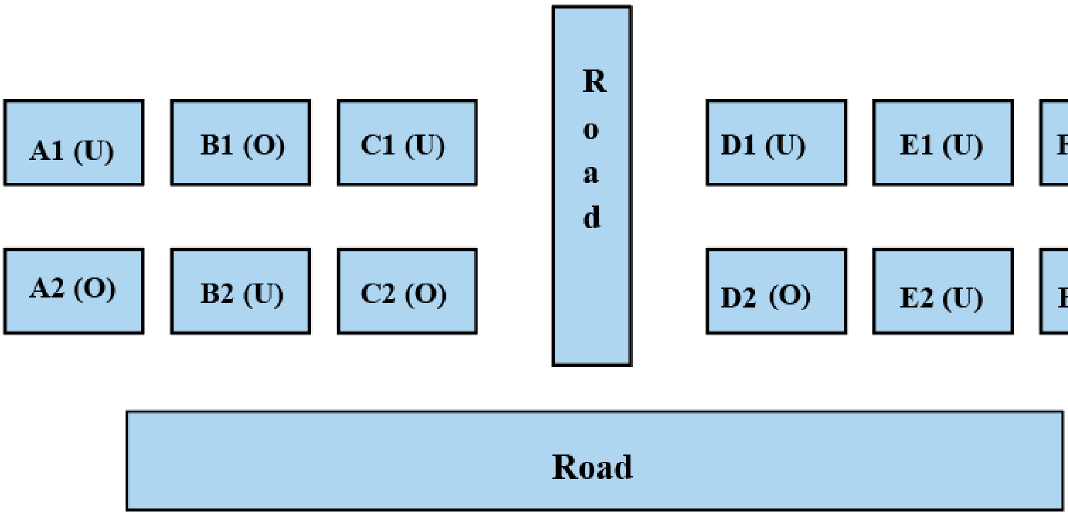
i) If the house has no working space, then: 10+5*0+3b = 15 => b = 5/3 (this is not possible since b has to be an integer value)

Hence, the house has parking space => 12+5*0+3b = 15 => b = 1 => One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10(there is no parking space)+ (5*0) + (3*the number of neighbours)
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.
Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.
Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:

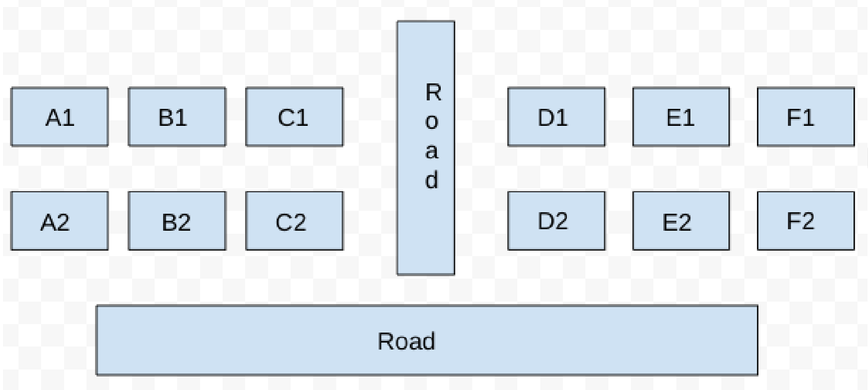


From the diagram, the vacant house with the maximum possible quoted price in column E is E2 when F2 is occupied.

The maximum possible quoted price of E2 is 10+5*1+3*2 = 21 Lacs. (E2 has no parking space because E1 has the parking space, and it is given that there is only one house with parking space in Block YY.)

VIDEO SOLUTION

93. **A**



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + 5 × (road adjacency value) + 3 × (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

=> 12+5a+3b = 24 => 5a+3b = 12 (a = road adjacency value, b= neighbor count)

The only value for which the equation satisfies is (a = 0, and b=4). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

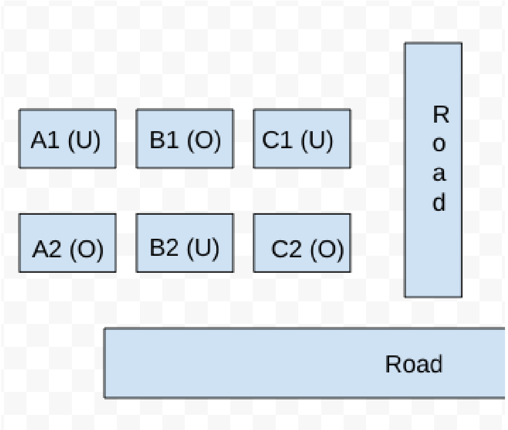
=> 10+5a+3b = 24 => 5a+3b = 14

=> (a, b) = (1, 3)

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and U = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: 12+5*1+3*b = 15 => 3b = -2 (which is not possible)

Hence, the house has no parking space => 10+5*1+3b = 15 => b = 0

b = 0 implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

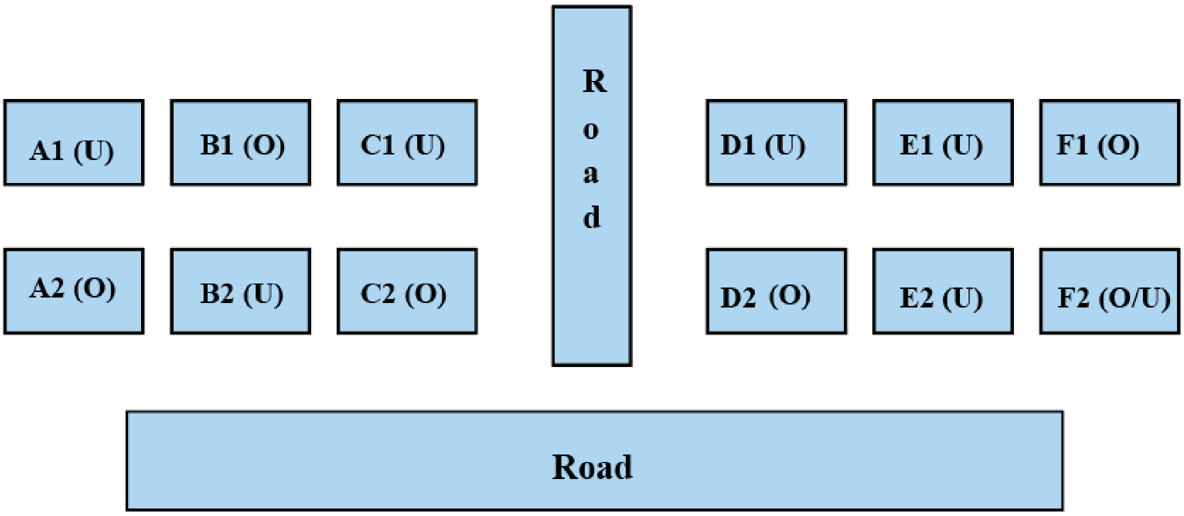
i) If the house has no working space, then: 10+5*0+3b = 15 => b = 5/3 (this is not possible since b has to be an integer value)

Hence, the house has parking space => 12+5*0+3b = 15 => b = 1 => One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be 10(there is no parking space)+ (5*0) + (3*the number of neighbours)
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.
Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.
Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:



From the diagram, we can see that E1 has the parking space (case 2).

The correct option is A

VIDEO SOLUTION

94. **C**

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10.(Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = 8/2 = 4

Also, the number of pumps that have medium contamination level = 20 - 8 - 4 = 12

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

We know that medium contamination level was recorded at exactly 8 pumps. Hence, option C is the correct answer.

VIDEO SOLUTION

95. **B**

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10.(Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = 8/2 = 4

Also, the number of pumps that have medium contamination level = 20 - 8 - 4 = 12

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

We can see that in case 1 the contamination level at P15 was recorded as medium. Let us check all the option one by one.

(Option :A) Contamination levels at P13 and P17 were recorded as the same. From the table, we can see that the contamination levels at P13 and P17 were recorded as medium. Hence, we can say that this statement is correct.

(Option :B) Contamination levels at P11 and P16 were recorded as the same. From the table, we can see that the contamination levels at P11 was recorded as Medium whereas at P16 it was recorded as Low. Hence, we can say that this statement is incorrect. Thus, option B is the correct answer.

VIDEO SOLUTION

96. **D**

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10.(Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = 8/2 = 4

Also, the number of pumps that have medium contamination level = 20 - 8 - 4 = 12

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Let us check the options one by one.

(Option:A) The contamination level at P20 was recorded as medium. This need not be true as we can see that in Case 1 at P20 low contamination level is recorded.

(Option:B) The contamination level at P13 was recorded as low. This need not be true as we can see that in Case 2(a), at P13 high contamination level is recorded.

(Option:C) The contamination level at P12 was recorded as high. This need not be true as we can see that in Case 2(a), at P12 medium contamination level is recorded.

(Option:D) The contamination level at P10 was recorded as high.. This is true for all cases. Hence, we can say that option D is the correct answer.

VIDEO SOLUTION

97. **D**

Let us draw the table and fill all absolute information present.

Pumps	Contamination level
P1	
P2	
P3	
P4	
P5	
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

In statement 3, it is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded.

In statement 1, it is given that contamination levels at three pumps among P1 - P5 were recorded as high. This is only possible when pumps 1, 3 and 5 have high level of contamination. Also, P6 was the only pump among P1 - P10 where the contamination level was recorded as low. Therefore, we can say that pumps 2 and 4 have medium level of contamination.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	
P8	
P9	
P10	
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that High contamination was recorded in only first 15 pumps. Therefore, we can say that the maximum number of pumps that can have high contamination level is '8'. (Consecutive pumps don't have same contamination level except one case)

Also, it is given that the number of pumps where high contamination levels were recorded was twice the number of pumps where low contamination levels were recorded. Hence, we can say that the number of pumps that have high contamination level is an even number less than or equal to '8'.

If the number of high contamination level pumps is '6', then there will be only '3' pumps with low contamination level. Consequently, we will need 11 (20 - 6 - 3) pumps with medium contamination level which is not possible since the number of pumps of a single type can't exceed 10.(Consecutive pumps don't have same contamination level except one case)

Therefore, we can say that the number of pumps that have high contamination level = 8

The number of pumps that have low contamination level = 8/2 = 4

Also, the number of pumps that have medium contamination level = 20 - 8 - 4 = 12

It is given that P7 and P8 were the only two consecutively numbered pumps where the same levels of contamination were recorded. If P7 and P8 recorded medium contamination level then there can be at max 7 pumps (P1, P3, P5, P9, P11, P13, P15) with high contamination level. Hence, we can say that pumps P7 and P8 recorded High contamination level. Therefore, we can uniquely determine the contamination level till P10.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	
P12	
P13	
P14	
P15	
P16	
P17	
P18	
P19	
P20	

It is given that High contamination levels were not recorded at any of the pumps P16 - P20. Therefore, we can say that these 5 pumps recorded low and medium contamination level. There are two cases possible.

Case 1: When there were 3 Low and 2 Medium contaminated level recorded in pumps P16 - P20.

3 Low contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Medium
P16	Low
P17	Medium
P18	Low
P19	Medium
P20	Low

Case 2: When there were 2 Low and 3 Medium contaminated level recorded in pumps P16 - P20.

3 Medium contamination level must have recorded in P16, P18 and P20. We can fill the table as follows.

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Low
P12	Medium
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	Low
P13	High
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Low
P14	Medium
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	Low
P15	High
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

Pumps	Contamination level
P1	High
P2	Medium
P3	High
P4	Medium
P5	High
P6	Low
P7	High
P8	High
P9	Medium
P10	High
P11	Medium
P12	High
P13	Medium
P14	High
P15	Low
P16	Medium
P17	Low
P18	Medium
P19	Low
P20	Medium

We can see that in case 2(a) the contamination level at P11 was recorded as low. Let us check all the option one by one.

(Option : A) The contamination level at P12 was recorded as high. This statement is incorrect as we can see in the table, the contamination level at P12 was recorded as medium.

(Option : B) The contamination level at P15 was recorded as medium. This statement is incorrect as we can see in the table, the contamination level at P15 was recorded as High.

(Option : C) The contamination level at P18 was recorded as low. This statement is incorrect as we can see in the table, the contamination level at P18 was recorded as Medium.

(Option : D) The contamination level at P14 was recorded as medium. This statement is correct as we can see in the table, the contamination level at P14 was recorded as Medium. Hence, we can say that option D is the correct answer.

VIDEO SOLUTION

98. **D**

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

5x4=20, 10x3=30, 15x2=30

The total possible with considering the minimum number of questions of each type = 20+30+30=80 marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = 9+4 =13 questions

2) {5,5,10} = 9+3 =12 questions

3) {10,10} = 9+2 =11 questions

4) {5,15} = 9+2 =11 questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is D

VIDEO SOLUTION

99. **A**
All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

5x4=20, 10x3=30, 15x2=30

The total possible with considering the minimum number of questions of each type = 20+30+30=80 marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = 9+4 =13 questions

2) {5,5,10} = 9+3 =12 questions

3) {10,10} = 9+2 =11 questions

4) {5,15} = 9+2 =11 questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is A

VIDEO SOLUTION

100. **D**

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

5x4=20, 10x3=30, 15x2=30

The total possible with considering the minimum number of questions of each type = 20+30+30=80 marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = 9+4 =13 questions

2) {5,5,10} = 9+3 =12 questions

3) {10,10} = 9+2 =11 questions

4) {5,15} = 9+2 =11 questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is D

VIDEO SOLUTION

101. **D**

All six sections had a common midterm (MT) and a common end term (ET) worth 100 marks each.

Each of MT and ET had at least four questions that were worth 5 marks, at least three questions that were worth 10 marks, and at least two questions that were worth 15 marks.

5x4=20, 10x3=30, 15x2=30

The total possible with considering the minimum number of questions of each type = 20+30+30=80 marks.

Rest 20 marks are possible by the following cases: {5,5,5,5} {5,5,10} {10,10} {5,15}

ET contained more questions than MT.

Thus MT cannot consider the case {5,5,5,5}

The number of questions in each case:

1) {5,5,5,5} = 9+4 =13 questions

2) {5,5,10} = 9+3 =12 questions

3) {10,10} = 9+2 =11 questions

4) {5,15} = 9+2 =11 questions

Considering MT and ET together, each faculty member prepared the same number of questions. The total number of questions should be multiple of 6, thus the total number of questions will be 24.

For ET and MT, there are 2 cases :

{5,5,5,5}{5,15}

{5,5,5,5}{10,10}

According to the statement (i), Annie prepared the fifth question for both MT and ET. For MT, this question carried 5 marks. Thus {10,10} case is not possible.

MT {5,5,5,5,5,10,10,10,15,15,15}

ET {5,5,5,5,5,5,5,5,10,10,10,15,15}

From statement (i),(ii),(iv),(v), every other faculty member prepared two questions for MT.
we can create the following table:

	1	2	3	4	5	6	7	8	9	10	11
MT	5	5	5	5	5	10	10	10	15	15	15
	F	F	C	C	A	B	B	E	E	D	D

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D		C		A			E					F

{ Annie(A), Beti(B), Chetan(C), Dave (D), Fakir(F) }

There are 24 questions in total so each faculty will make 4 questions.

We can create the following table for ET.

	1	2	3	4	5	6	7	8	9	10	11	12	13
ET	5	5	5	5	5	5	5	5	10	10	10	15	15
	D	D	C	C	A	A	A	E	E	B	B	F	F

Hence the correct option is D

VIDEO SOLUTION

102. **C**

Considering (i), Ignesh has to eat 6 vadas, since 6 is the only multiple of 3.

Also, using the same information, we can say that a person consumes 2 vadas and 4 idlis.

Using (vii), Bimal eats 2 more idlis than Ignesh.

Possibilities:

Bimal - 8, Ignesh - 6

Bimal - 6, Ignesh - 4

But Ignesh cannot have 4 idlis because the person who eats 4 idlis eats 2 vadas.

Hence we take Bimal - 8 and Ignesh - 6.

Also, we get that Bimal eats 6 - 2 = 4 vadas.

So far, we get the following information.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4

Using (vi), there is a person who eats twice as many idlis as Mukesh. The only pair satisfying is 8, 4.

So, Mukesh eats 4 idlis. Plus the person who eats 4 idlis eats 2 vadas. Hence, we get

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2

Daljit also eats Vada as per info (v), so we get the following

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit		1
Sandip		0

(iv) gives us the information that the one who eats 1idli does not have vada.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit	5	1
Sandip	1	0

Considering the persons who had chutney and those who didn't, 3 persons do not have chutney. Bimal is one of them(the one eating 4 vadas).

Mukesh is the second one to not take chutney(last hint). Also, Sandip does not take chutney. Hence, we get this information as well.

	Idlis	Vadas	Chutney
Ignesh	6	6	☑
Bimal	8	4	~
Mukesh	4	2	~
Daljit	5	1	☑
Sandip	1	0	~

Hence, Ignesh eating 6 vadas and 6 idlis eat Chutney.

VIDEO SOLUTION

103. **A**

Considering (i), Ignesh has to eat 6 vadas, since 6 is the only multiple of 3.

Also, using the same information, we can say that a person consumes 2 vadas and 4 idlis.

Using (vii), Bimal eats 2 more idlis than Ignesh.

Possibilities:

Bimal - 8, Ignesh - 6

Bimal - 6, Ignesh - 4

But Ignesh cannot have 4 idlis because the person who eats 4 idlis eats 2 vadas.

Hence we take Bimal - 8 and Ignesh - 6.

Also, we get that Bimal eats 6 - 2 = 4 vadas.

So far, we get the following information.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4

Using (vi), there is a person who eats twice as many idlis as Mukesh. The only pair satisfying is 8, 4.

So, Mukesh eats 4 idlis. Plus the person who eats 4 idlis eats 2 vadas. Hence, we get

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2

Daljit also eats Vada as per info (v), so we get the following

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit		1
Sandip		0

(iv) gives us the information that the one who eats 1idli does not have vada.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit	5	1
Sandip	1	0

Considering the persons who had chutney and those who didn't, 3 persons do not have chutney. Bimal is one of them(the one eating 4 vadas).

Mukesh is the second one to not take chutney(last hint). Also, Sandip does not take chutney. Hence, we get this information as well.

	Idlis	Vadas	Chutney
Ignesh	6	6	☑
Bimal	8	4	~
Mukesh	4	2	~
Daljit	5	1	☑
Sandip	1	0	~

Option A, stating that Daljit eats 5 idlis, is right.

VIDEO SOLUTION

104. **C**

Considering (i), Ignesh has to eat 6 vadas, since 6 is the only multiple of 3.

Also, using the same information, we can say that a person consumes 2 vadas and 4 idlis.

Using (vii), Bimal eats 2 more idlis than Ignesh.

Possibilities:

Bimal - 8, Ignesh - 6

Bimal - 6, Ignesh - 4

But Ignesh cannot have 4 idlis because the person who eats 4 idlis eats 2 vadas.

Hence we take Bimal - 8 and Ignesh - 6.

Also, we get that Bimal eats 6 - 2 = 4 vadas.

So far, we get the following information.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4

Using (vi), there is a person who eats twice as many idlis as Mukesh. The only pair satisfying is 8, 4.

So, Mukesh eats 4 idlis. Plus the person who eats 4 idlis eats 2 vadas. Hence, we get

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2

Daljit also eats Vada as per info (v), so we get the following

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit		1
Sandip		0

(iv) gives us the information that the one who eats 1idli does not have vada.

	Idlis	Vadas
Ignesh	6	6
Bimal	8	4
Mukesh	4	2
Daljit	5	1
Sandip	1	0

Considering the persons who had chutney and those who didn't, 3 persons do not have chutney. Bimal is one of them(the one eating 4 vadas).

Mukesh is the second one to not take chutney(last hint). Also, Sandip does not take chutney. Hence, we get this information as well.

	Idlis	Vadas	Chutney
Ignesh	6	6	☑
Bimal	8	4	~
Mukesh	4	2	~
Daljit	5	1	☑
Sandip	1	0	~

Hence, Ignesh eats 6 Vadas.

VIDEO SOLUTION

105. **A**

Let a,d,j,g be the shots put by Ashish, Dhanraj, Ganesh and Jugraj respectively.

According to given conditions we have ,
$g = a-8$; $d + r = 37$; $j = d + 8$; $a = 5 + d$; $a + g = 40$

Solving these equations, we have $a = 24$, $d = 19$ and $j = 27$ and $r = 18$. Hence option A is the correct answer.

VIEW SOLUTION

106. **A**

Let a,d,j,g be the shots put by ashish,dhanraj,ganesh and jugraj respectively. According to given conditions we have ,
$g=a-8$;

$d+r=37$;

$j=d+8$;

$a=5+d$;

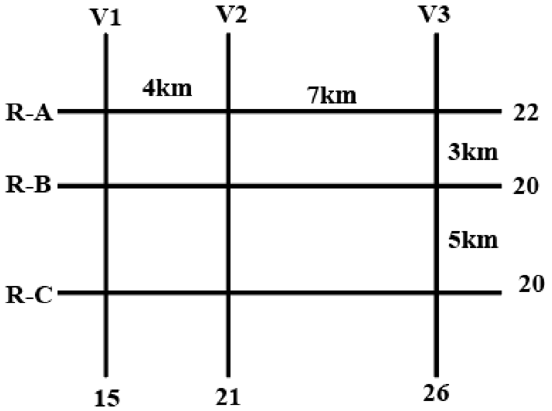
$a+g=40$.

Solving we have $a=24$, $d=19$ and $j=27$.so $d+j=45$

VIEW SOLUTION

107. **3**

This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.

We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.

This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.

Next piece of information that is given is that, road distance between the ATM with the second highest cashrequirement and the ATM located at the intersection of R-C and V3 is12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.

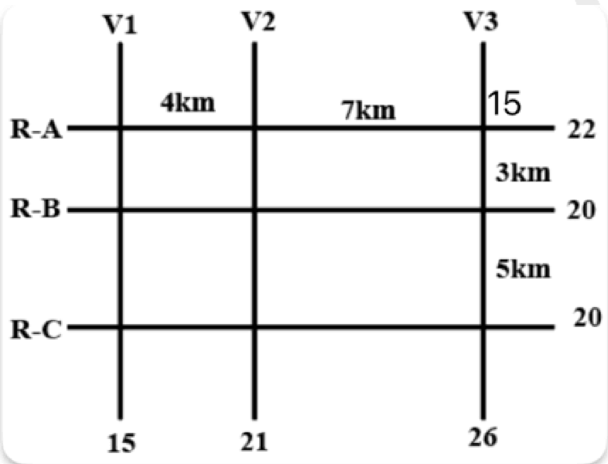
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.

There are only two possible choice, either RA or V3.

Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.

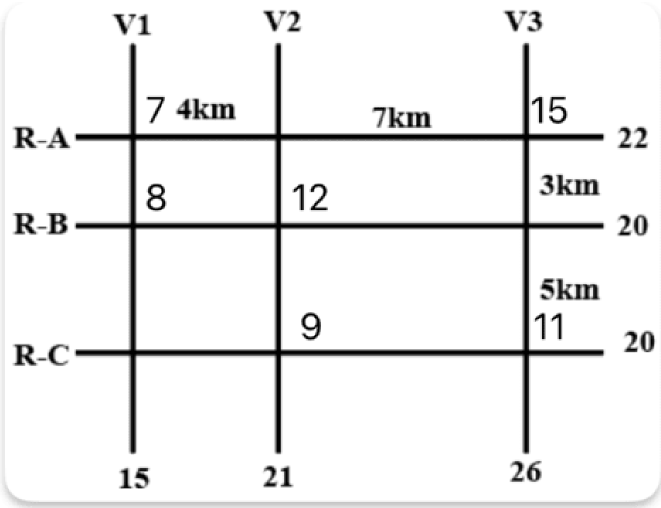
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)

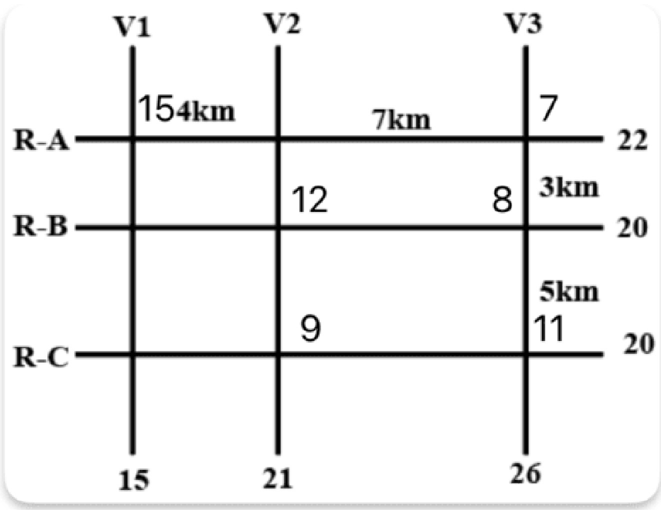


We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)
There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.

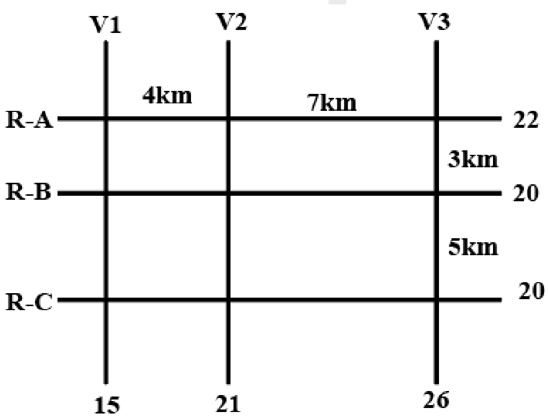


Using the two cases, we can answer the given questions.

Three ATM's with 10L or more: 15L, 12L and 11L. Answer is 3.

VIDEO SOLUTION

108. **B**
This is the figure that has been given to us,

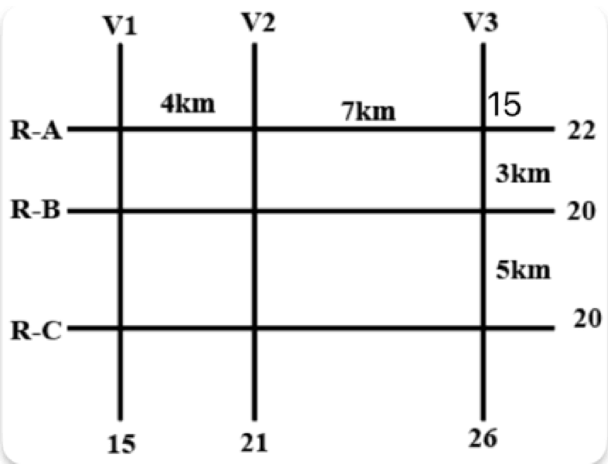


We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.
We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity

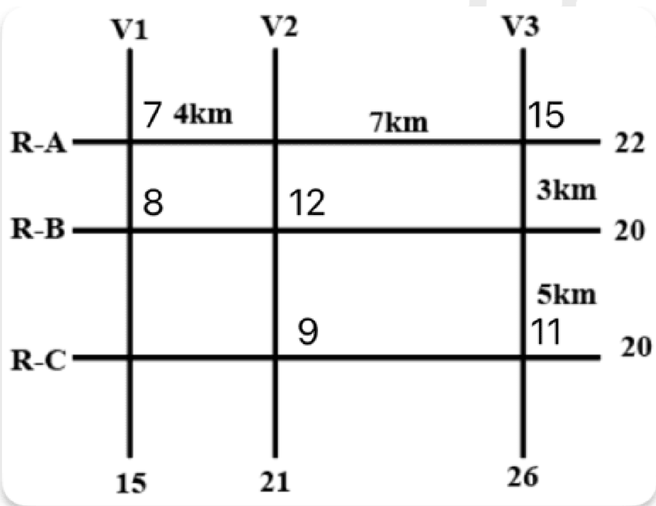
being 15L and lowest being 7L.
This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.
Next piece of information that is given is that, road distance between the ATM with the second highest cashrequirement and the ATM located at the intersection of R-C and V3 is12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.
There are only two possible choice, either RA or V3.
Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)

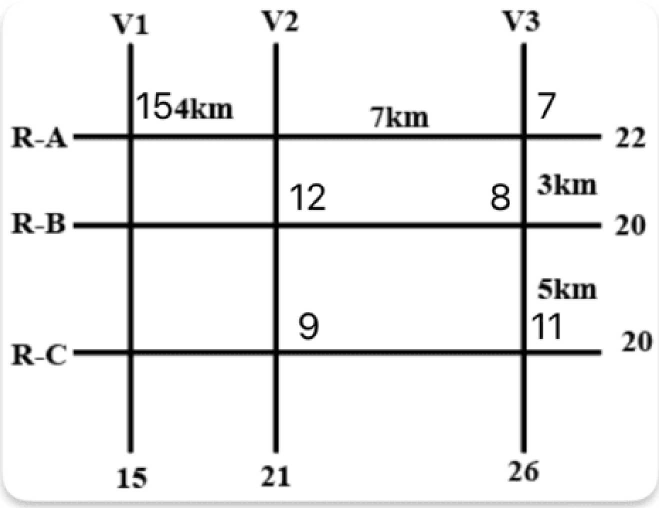


We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.
We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)
There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and

RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.

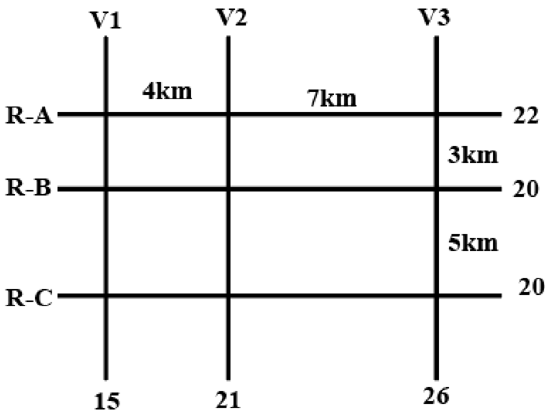


Using the two cases, we can answer the given questions.

The only statement that is correct considering both the cases is Option B
"The ATM placed at the (R-C, V2) intersection has a cash requirement of Rs. 9 Lakhs."

VIDEO SOLUTION

109. **A**
This is the figure that has been given to us,

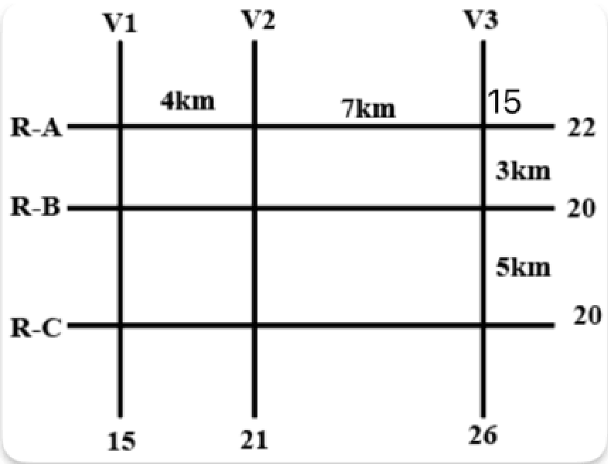


We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.
We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.
This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.
Next piece of information that is given is that, road distance between the ATM with the second highest cashrequirement and the ATM located at the intersection of R-C and V3 is12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

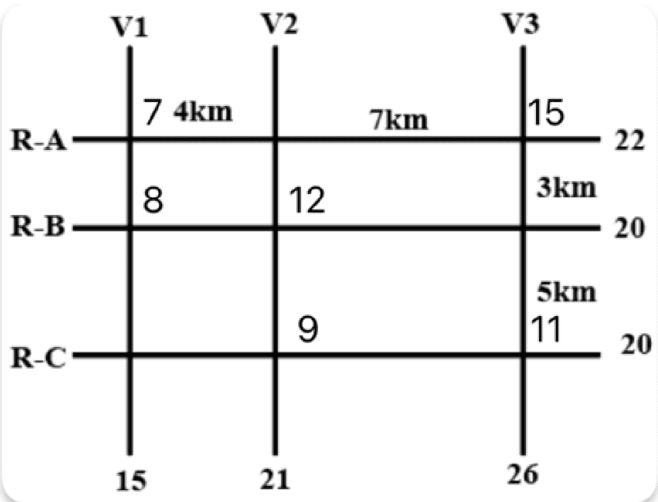
Now, let us start arranging the ATM's.
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.
There are only two possible choice, either RA or V3.
Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not

possible since there cannot be an ATM with 5L capacity.
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

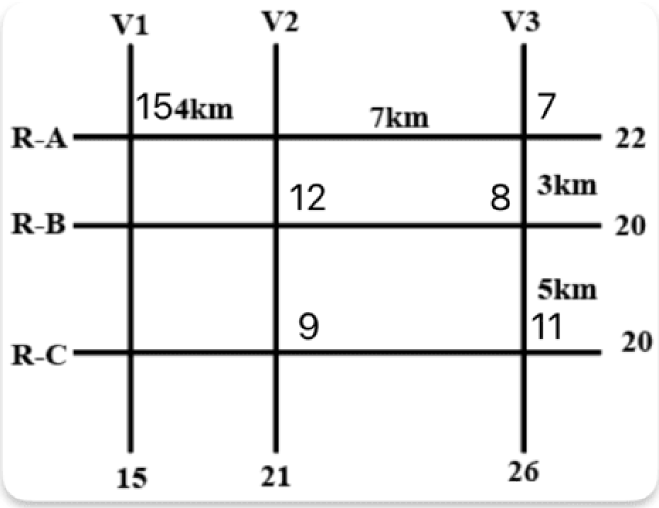
Case 1: 15L ATM is on the intersection (RA, V3)



We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.
We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)
There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



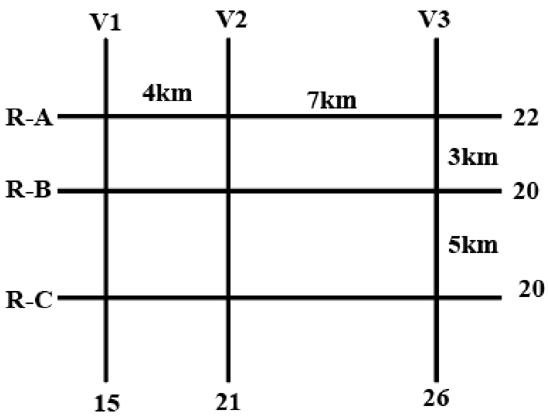
Using the two cases, we can answer the given questions.

We see that in both cases, RA RB and RC have two ATMs
Only in the first case V1 V2 and V3 have two ATM's, in the second case V3 has 3 ATMs and V1 has 1.

Hence only Statement A is correct.

VIDEO SOLUTION

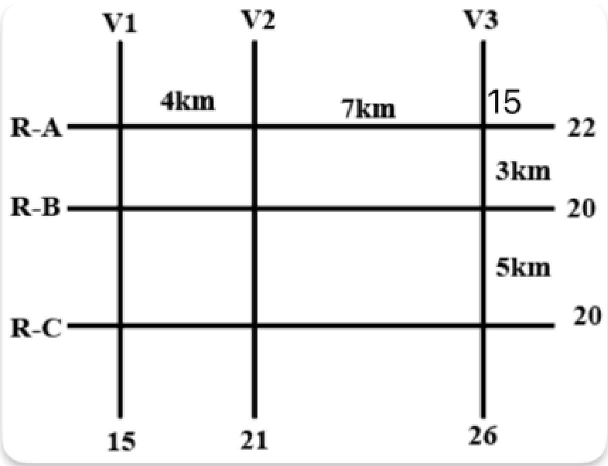
110. **D**
This is the figure that has been given to us,



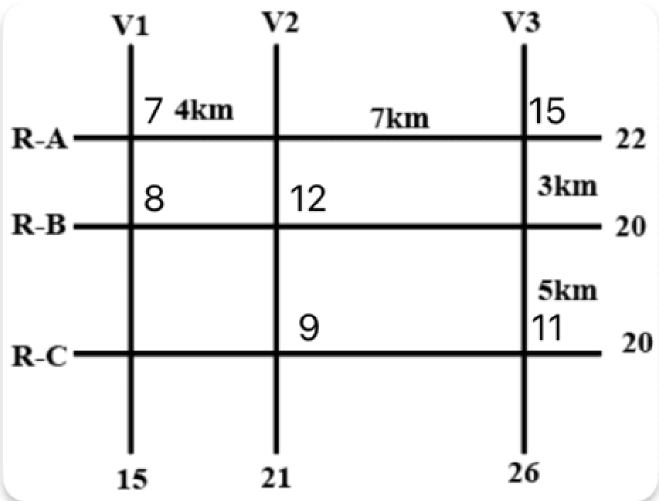
We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.
We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.
This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.
Next piece of information that is given is that, road distance between the ATM with the second highest cashrequirement and the ATM located at the intersection of R-C and V3 is12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

Now, let us start arranging the ATM's.
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.
There are only two possible choice, either RA or V3.
Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

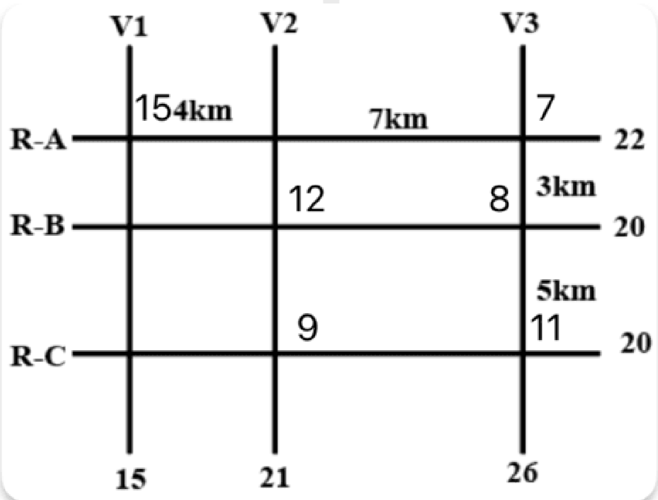
Case 1: 15L ATM is on the intersection (RA, V3)



We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.
We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)
There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



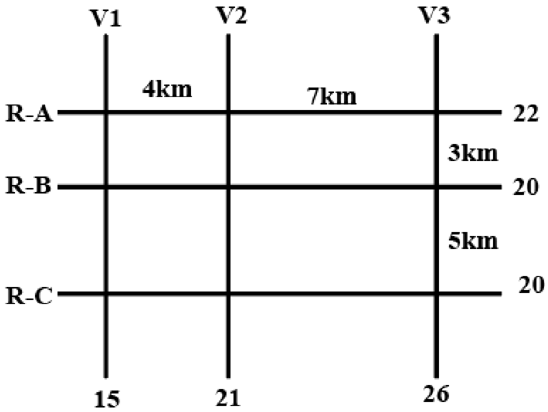
Using the two cases, we can answer the given questions.

Second highest is 12L and second lowest is 8L

In Case 1 the distance between 12L and 8L ATM is 4km and in Case 2 the distance between 12L and 8L ATM is 7km.
Hence the answer is either 4km or 7km.

VIDEO SOLUTION

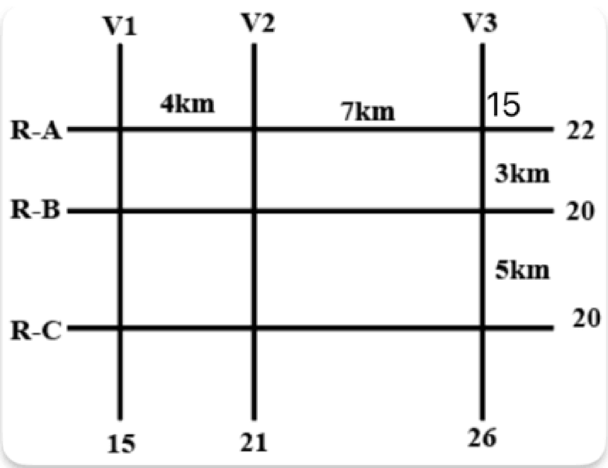
111.**3**
This is the figure that has been given to us,



We are given the information that, out of the 9 intersections in the figure, 6 of them have ATMs. That means, 3 of these intersections are empty.
We are also told that, the ATMs with the highest and lowest capacity are on the same road, highest capacity being 15L and lowest being 7L.
This information not only gives us clues about the location of these two ATMs but also, now we know the upper and lower bounds for cash in the six ATMs with distinct cash.
Next piece of information that is given is that, road distance between the ATM with the second highest cashrequirement and the ATM located at the intersection of R-C and V3 is12 km. Since we can only traverse on the roads, from (RC, V3) we have to either traverse the 5km road or the 7km road. The only way it can add up to 12 is 5+7. That means, ATM with the second highest capacity is at (RB, V2).

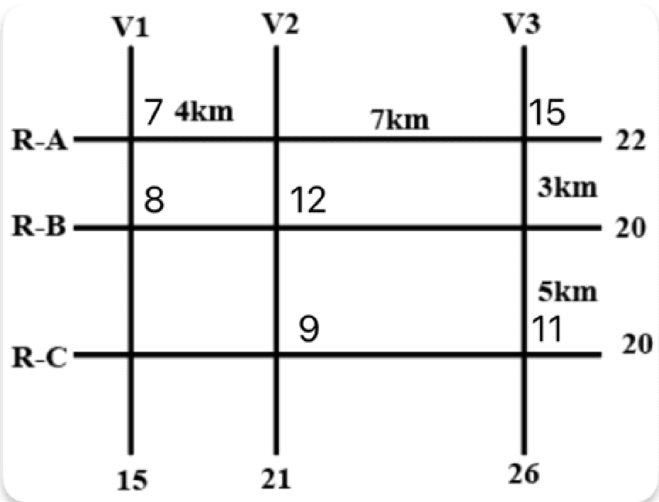
Now, let us start arranging the ATM's.
We are told that 15 and 7 are on the same road. Since we are given the total capacities on the roads, we need to identify the roads with capacity higher or equal to 22.
There are only two possible choice, either RA or V3.
Looking at V3, we see that 15L ATM cannot come at (RB, V3) or (RC, V3) since the RB and RC capacity is 20, and the minimum ATM limit is 7L, if a 15L ATM is on a road with total capacity 20L, this is a situation that is not possible since there cannot be an ATM with 5L capacity.
The same is the case with the intersection (RA, V2). So, we can narrow down the fact that the 15L ATM has to either be at (RA, V1) or (RA, V3)

Case 1: 15L ATM is on the intersection (RA, V3)



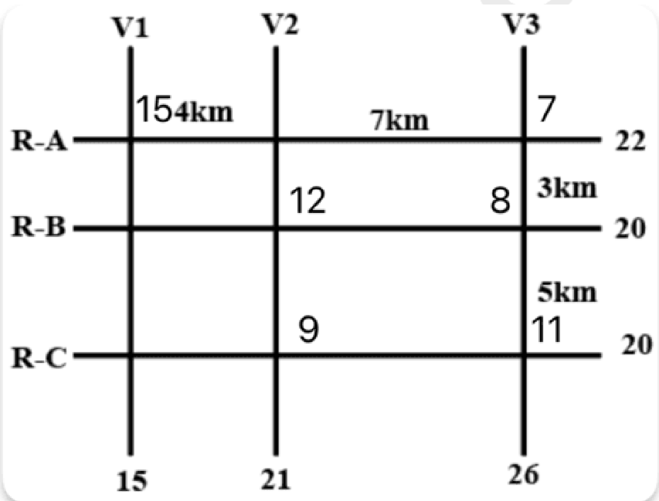
We see that, for V3 to add upto 26, there has to be an ATM with cash of 11L, there cannot be two ATM's since the minimum capacity is 7L.

We can place the 11L ATM at (RB, V3) or (RC, V3), if we place them at either of these intersections, the remaining ATM has to have a capacity of 9L for the same reason. 9L cannot be at (RC, V1) or (RB, V1) since the total capacity of V1 is 15L and there cannot be an ATM with 6L. And it also cannot be at (RB, V2) since there is already an ATM with 11L that means the ATM with the second highest capacity cannot be 9. So that means 9L has to be at (RC, V2). And then filling in the rest of the numbers we get the final arrangement for Case-1.



Case 2: When 15L is at (RA, V1)

There can be no other ATM on V1 in this scenario, 7L ATM which is on RA cannot be on (RA, V2) considering the sum of the numbers on V2 is 21, and there cannot be 7+7 or a 14L ATM since the capacity of both RB and RC is 20. So, 7L has to be on V3, and since there cannot be a single ATM of 19L on V3, there has to be two other ATMs on V3 adding up to 19. Rearranging the numbers, we get the scenario for the second case.



Using the two cases, we can answer the given questions.

ATMs that can be uniquely determines are the ATMs with cash 9L, 11L and 12L. Hence the answer is 3.

VIDEO SOLUTION

112. **D**

Let's look at option 4.

Order of cars is 8,2,3,V,5,6,7. This sequence is easily possible.

Let's say cars 1,2,3,4,5,6,7 arrive one after the another.

Now Car 1 leaves and Car 8 takes that place.

Finally Car 4 leaves. Hence we can see that this combination of cars is possible

VIDEO SOLUTION

113. **C**

The original sequence as given in the question is 4,5,6,V,3

This is possible when cars 1,2,3 arrived and then cars 1 and 2 leave. After that cars 4,5 and 6 arrive.

Now there are 4 slots to the left of car 3. This is only possible when cars 1 and 2 were SUVs. Now out of these 4 slots,

3 slots are occupied by cars 4,5 and 6. As a result these are compact cars . Car 3 can be a SUV or a Compact car and it won't impact the final solution.

VIDEO SOLUTION

114. **D**

Here we can see that cars 3 and 5 are still in their position. Thus car 4 was not the first car to leave, either 1 or 2 left before 4. Let's say only car 2 left before car 4. Now supposingly if car 2 is an SUV, car 6 was parked in that lot. Thus car 2 and car 7 are compact cars. Option 4 is correct.

VIDEO SOLUTION

115. **D**

following is the order of arrival and departure of cars

	1 (SUV)	2	3	4 (SUV)	
Car 1 leaves	V V	2	3	4 (SUV)	
Car 5 Arrives	5 V	2	3	4 (SUV)	
Car 6 (SUV) Arrives	5 V	2	3	4 (SUV)	6 (SUV)
Car 4 leaves	5 V	2	3	V V	6 (SUV)
Car 7 (SUV) Arrives	5 V	2	3	7 (SUV)	6 (SUV)
Car 8 Arrives	5 8	2	3	7 (SUV)	6 (SUV)

As we can see that car 2 and car 7 are parked next to car 3

VIDEO SOLUTION

116. **6**

To solve this question we can use the answer of the previous question, since maximum 9 red beads are possible, filling the remaining space with green and blue beads, in such a way that number of blue beads is minimised

R	G	B	R	G
G	R	G	B	R
B	G	R	G	B
R	B	G	R	G
G	R	B	G	R

Hence number of blue beads is 6

VIDEO SOLUTION

117. **6**

6 more beads can be placed as shown

R			R	
		R		
	R			R
R			R	
		R		

VIDEO SOLUTION

118. **9**

R			R	
	R			R
		R		
R			R	
	R			R

Maximum 9 red beads are possible as shown here

VIDEO SOLUTION

119. **2**

Since we are required to use only two colours, these can be either
1. Green + Blue
2. Blue + Red
3. Green + Red

But we know that Between any two Red coloured beads in a row or a column, there have to be both a Green and a Blue coloured bead. Hence without both Green and Blue, Red beads cannot be used to fill the grid.Thus if we use only Green and Blue beads, the two configurations that are possible are:

G	B	G	B	G
B	G	B	G	B
G	B	G	B	G
B	G	B	G	B
G	B	G	B	G

There are only 2 configurations possible

B	G	B	G	B
G	B	G	B	G
B	G	B	G	B
G	B	G	B	G
B	G	B	G	B

VIDEO SOLUTION

120. **B**

The different possible cases are:

	Amar	Akbar	Antony
1	Green	Red	Red
2	Blue	Red	Red
3	Green	Red	Green
4	Green	Blue	Green
5	Blue	Blue	Green
6	Blue	Blue	Red

From the table we can see among all of the statements, only fourth statement is false, as if amar wears green, anthony has to wear green.

VIEW SOLUTION

121. **A**
Akbar can wear either red or blue.
Anthony can wear either red or green.
So when both of them wear same colour that has to be red.
And Amar can wear either blue or green.
Hence our answer will be A.

VIEW SOLUTION